

NATIONAL HIGHER SCHOOL OF MATHEMATICS
المدرسة الوطنية العليا في الرياضيات

CHAPTER 1

Counting

Discrete Mathematics 1 | First year

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Roadmap of the chapter

1. Introduction: what is discrete mathematics?
2. Fundamental principles: product or sum?
3. Distinguishability, ordering and repetition
4. Arrangements and permutations
5. Combinations, binomial theorem, Pascal's triangle
6. Multinomial coefficients and theorem
7. Combinations with repetition

Convention for this course

Statements and examples are on the slides; **proofs are done on the board**

✎ **Proof on the board**

01

Introduction

What is discrete mathematics, and why do we count?

What is Discrete Mathematics?

Definition

Discrete Mathematics is a field of mathematics that deals with distinct and countable objects, rather than quantities that change continuously.

- It studies structures that are fundamentally discrete: numbers, logical statements, algebraic structures, ...
- It forms the backbone of **computer science**, **cryptography**, and many branches of **pure mathematics**.

Key concepts in Discrete Mathematics

- **Sets:** collections of distinct objects.
- **Logic:** principles of reasoning and valid arguments.
- **Algebraic structures:** groups, rings, fields.
- **Number theory:** properties of the integers.
- **Graph theory:** vertices joined by edges; networks, relations, algorithms.

Combinatorics

The art of **counting, organizing and selecting** objects. We focus on enumerative combinatorics. Key questions:

- Is the configuration possible?
- In how many ways can it be made?
- How do we find such a configuration?

Some questions in combinatorics

- How many distinct ways can the letters of the word "NHSM" be arranged?
- 12 students, 3 different prizes (gold, silver, bronze): in how many ways can the prizes be distributed?
- How many ways can you arrange 6 balls into 4 boxes?
- How many ways can you assign 5 distinct colors to 5 identical chairs in a row?
- How many passwords of length n are possible using 26 letters?
- 7 tasks, 3 employees: in how many ways can the tasks be assigned if each employee gets at least one task?

Listing versus counting

Remark

When doing enumeration, it is very important to understand the distinction between listing all the configurations and determining their number.



Why study combinatorics?

- **Computer science:** efficient algorithms, data structures, cryptographic systems, programming, network theory.
- **Engineering:** optimizing system designs, network configurations, resource allocations.
- **Mathematics:** number theory, algebraic structures, discrete optimization.
- **Statistics:** the foundation of probability theory and statistical models.
- **Physics:** quantum mechanics and statistical physics, where discrete models are essential.

Powerful techniques for logical reasoning, problem-solving and analyzing complex systems.

A first exercise

✎ Exercise

How many different 2-letter words, including nonsense words, can be produced using the letters $\{H, I, M\}$ (letters may be repeated)?

A first exercise: solution

Solution

The set of all possible 2-letter words, including nonsense words, is

$$\{HI, HM, IM, IH, MH, MI, II, HH, MM\};$$

there are 9 such words.

Let us change the question a little

✎ **Exercise**

How many different 26-letter words, including nonsense words, can be produced using the letters of the alphabet (letters may be repeated)?

Let us change the question a little: solution

Solution

There are 26^{26} such words. Even using a computer that prints 1 billion words per second, it will take about

$$1.95 \times 10^{18} \text{ centuries}$$

to print all the possible words.

02

Fundamental Principles of Counting

Product or sum?

Cardinality of a set

Definition

The **cardinality** of a finite set S is the number of its elements. It is denoted by $|S|$.

Examples

$$S = \{a, b, c, d\} \implies |S| = 4, \qquad T = \{1, 2, 3, 4, 5, 6, 7\} \implies |T| = 7.$$

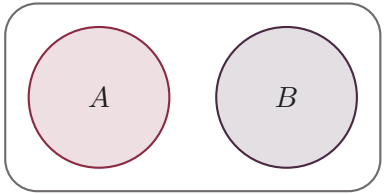
Addition principle

Theorem (Addition Principle)

If A and B are two **disjoint** sets ($A \cap B = \emptyset$), then

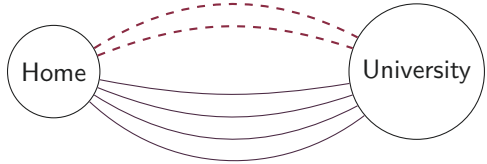
$$|A \cup B| = |A| + |B|.$$

✎ **Proof on the board**



Addition principle: examples

1. 2 different train routes and 4 different bus routes from home to the university.
   With $A =$ train routes, $B =$ bus routes: $|A| + |B| = 6$ possible routes.



2. A class has 8 boys and 5 girls; we choose one representative:
   $|\text{Boys}| + |\text{Girls}| = 13$ possible choices.

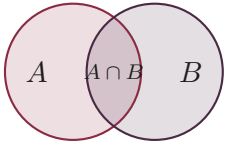
Disjointness is essential

Remark

If $A \cap B \neq \emptyset$, the elements of $A \cap B$ are counted twice, and

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

This is the starting point of the **Inclusion–Exclusion Principle**.



Generalised addition principle

Corollary (Generalised Addition Principle)

If $A_1, A_2, \ldots, A_n$ are n pairwise disjoint sets ($A_i \cap A_j = \emptyset$ for all $i \neq j$), then

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_{i=1}^{n} |A_i|.$$

✎ **Proof by induction on the board**

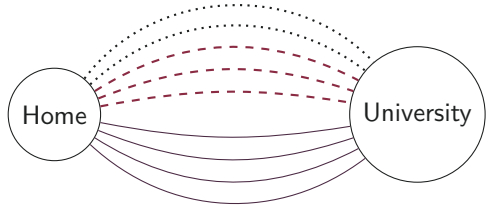
Generalised addition principle: example

Three modes of transportation:

- 3 train routes (A_1), dashed,
- 4 bus routes (A_2), solid,
- 2 tram routes (A_3), dotted.

The sets are pairwise disjoint (one mode at a time), so

$$|A_1 \cup A_2 \cup A_3| = 3 + 4 + 2 = 9.$$



Addition principle: alternate form

Theorem (Addition Principle: Alternate Form)

If a problem consists of solving **exactly one** of k tasks, where no way of doing one task is also a way of doing another, and each task i $(1 \leq i \leq k)$ can be done in n_i ways, then the total number of solutions is

$$\sum_{i=1}^{k} n_i = n_1 + n_2 + \cdots + n_k.$$

Multiplication principle

Definition (Cartesian product)

$A \times B$ is the set of all ordered pairs (a, b) with $a \in A$ and $b \in B$.

Theorem (Multiplication Principle)

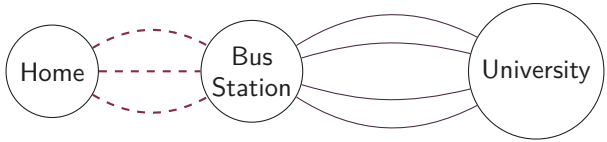
If A and B are two sets, then

$$|A \times B| = |A| \cdot |B|.$$

✎ **Proof on the board**

Multiplication principle: examples

1. Choose one of 3 train routes from home to the bus station, **then** one of 4 bus routes from the bus station to the university: $3 \times 4 = 12$ routes.



2. 8 boys and 5 girls; a committee of one boy **and** one girl: $8 \times 5 = 40$ ways.

Generalised multiplication principle

Corollary (Generalised Multiplication Principle)

If $A_1, A_2, \ldots, A_n$ are n sets, then

$$|A_1 \times A_2 \times \cdots \times A_n| = |A_1| \cdot |A_2| \cdots |A_n|.$$

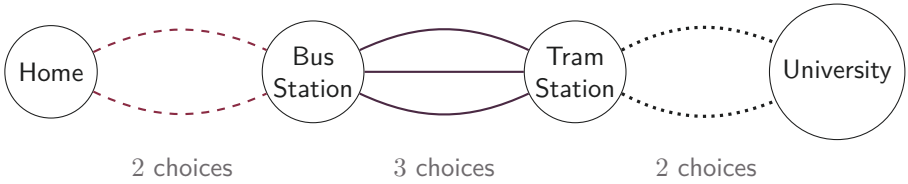
✎ **Proof by induction on the board**

Example

Home $\rightarrow$ bus station: one of 2 train routes; bus station $\rightarrow$ tram station: one of 3 bus routes; tram station $\rightarrow$ university: one of 2 tram routes.

$$|A_1 \times A_2 \times A_3| = 2 \times 3 \times 2 = 12 \text{ ways.}$$

A journey in three stages



Total: $2 \times 3 \times 2 = 12$ different ways.

Multiplication principle: alternate form

Theorem (Multiplication Principle: Alternate Form)

If a problem consists of solving k tasks **sequentially**, and each task i $(1 \leq i \leq k)$ can be performed in n_i ways, then the total number of solutions is

$$\prod_{i=1}^{k} n_i = n_1 \times n_2 \times \cdots \times n_k.$$

Remark

The **number** n_i of ways to perform task i must not depend on the choices made in the previous tasks (the choices themselves may change, but not their number).

Product or sum?

Rule of thumb

"**or**" (one case among disjoint cases) $\Longrightarrow$ add; "**and then**" (successive steps) $\Longrightarrow$ multiply.

Example

How many passwords of length 1, 2 or 3 can be formed with 26 letters?

A password has length 1 **or** 2 **or** 3 (addition), and a password of length i is i successive choices (multiplication):

$$26 + 26^2 + 26^3 = 18\,278.$$

03

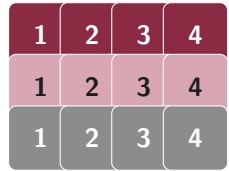
Distinguishability, Ordering & Repetition

Three questions to ask before counting anything

Distinguishability

Definition (Distinguishability)

Two elements of S are **indistinguishable** if they are identical with respect to the property in question; otherwise they are **distinguishable**.

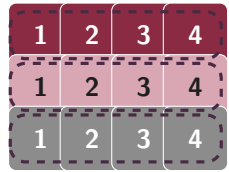


Distinguishability

Definition (Distinguishability)

Two elements of S are **indistinguishable** if they are identical with respect to the property in question; otherwise they are **distinguishable**.

- **By color:** 3 distinguishable groups, each with 4 indistinguishable squares.



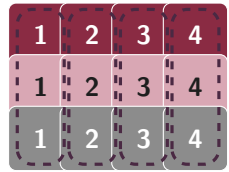
3 groups (colors)

Distinguishability

Definition (Distinguishability)

Two elements of S are **indistinguishable** if they are identical with respect to the property in question; otherwise they are **distinguishable**.

- **By color:** 3 distinguishable groups, each with 4 indistinguishable squares.
- **By number:** 4 distinguishable groups, each with 3 indistinguishable squares.



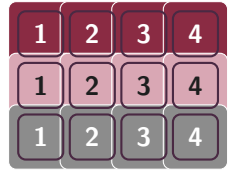
4 groups (numbers)

Distinguishability

Definition (Distinguishability)

Two elements of S are **indistinguishable** if they are identical with respect to the property in question; otherwise they are **distinguishable**.

- **By color:** 3 distinguishable groups, each with 4 indistinguishable squares.
- **By number:** 4 distinguishable groups, each with 3 indistinguishable squares.
- **By color and number:** 12 pairwise distinguishable squares.



12 different squares

Ordering

Definition (Ordering)

If the selected elements are treated equally and play the same role, the selection is **unordered**. If they can occupy different roles (their positions matter), the selection is **ordered** (a list).

Example

The 100-meter final at the Olympics, with 8 participants:

- select 3 runners for a doping test: unordered;
- select 3 runners for gold, silver and bronze medals: ordered, since the positions matter.

If the items are truly indistinguishable, then "order" becomes meaningless!

Ordered or unordered? A picture

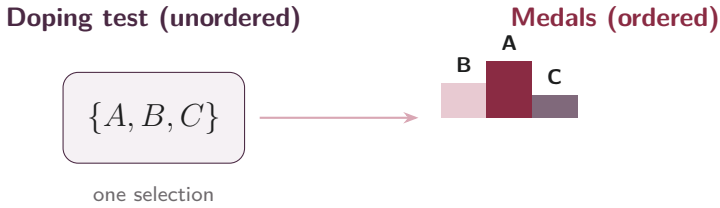
Doping test (unordered)

$\{A, B, C\}$

one selection

Medals (ordered)

Ordered or unordered? A picture



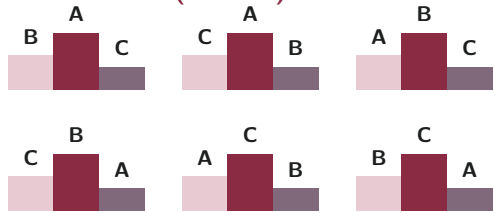
Ordered or unordered? A picture

Doping test (unordered)

$\{A, B, C\}$

one selection

Medals (ordered)



$3! = 6$ ordered results for **one** unordered selection

Repetition

Definition (Repetition)

If an element of S can appear more than once in the selection, **repetition is allowed**. If each element can appear only once, repetition is **not allowed**.

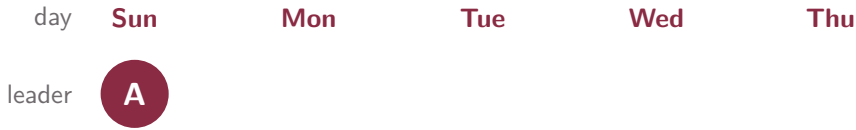
Example

We select a leader from a group of students **each day**. Repetition is possible: a student can be chosen more than once.

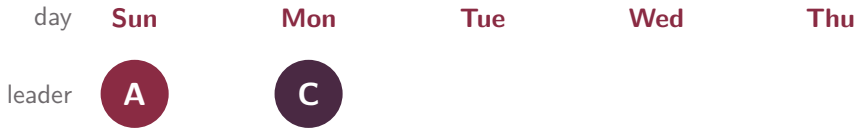
Does the order matter in this case?

We now consider selections of k elements under the conditions of **ordering** and **repetition**.

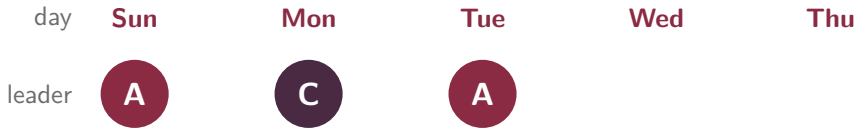
Repetition: a week of leaders



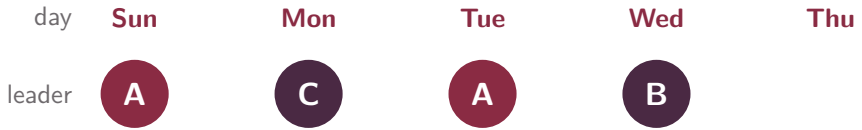
Repetition: a week of leaders



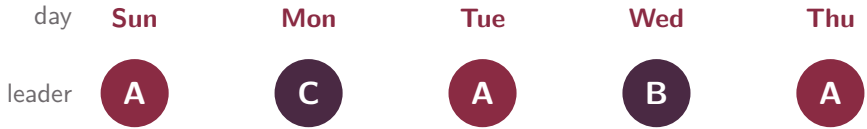
Repetition: a week of leaders



Repetition: a week of leaders



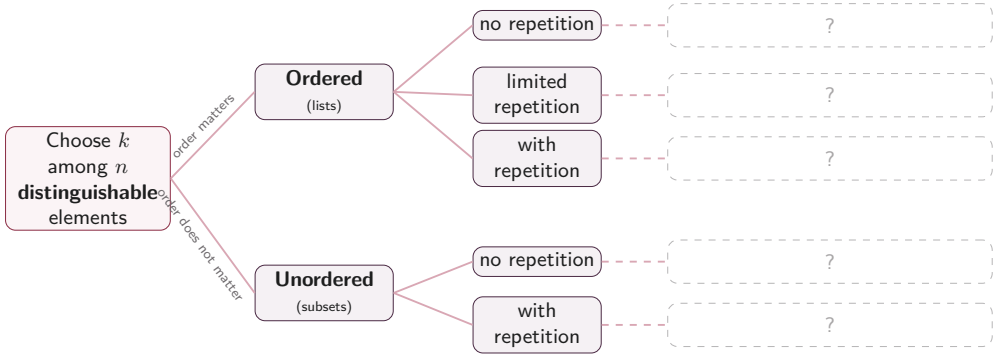
Repetition: a week of leaders



Student A is chosen **three times**: repetition is allowed.
The days are different positions: the order matters.

Where are we? The tree of notions

Choose k among n distinguishable elements: **does the order matter? Is repetition allowed?**



3.1

Arrangements & Permutations

Ordered selections, with or without repetition

Arrangements (k-permutations)

Definition

Let $S = \{e_1, \ldots, e_n\}$. An **arrangement** (or k-**permutation**) of k elements from S is any **ordered** sequence $(a_1, a_2, \ldots, a_k)$ of k elements selected from S.

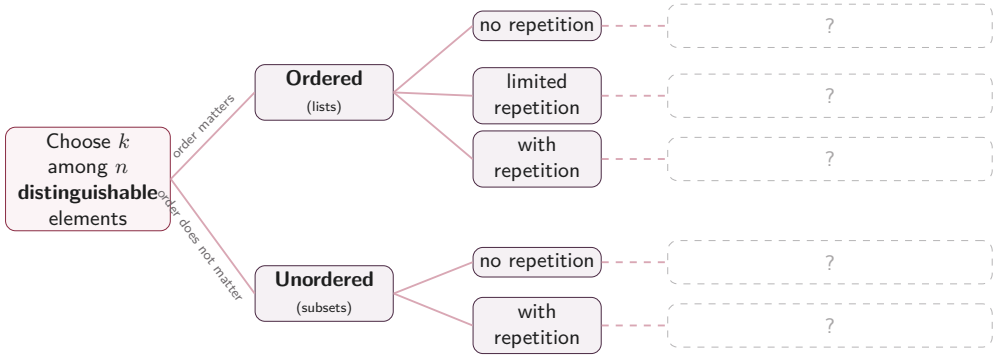
Examples

- $S = \{1, 2, \ldots, 9\}$: $123,\ 112,\ 344,\ 789,\ 258$ are arrangements of 3 elements.
- $S = \{A, B, C, D, E, F\}$: $ABC,\ DEF,\ AAB,\ CDC,\ CDA$ are arrangements of 3 elements.

*An arrangement can simply be described as a **listing** of elements.*

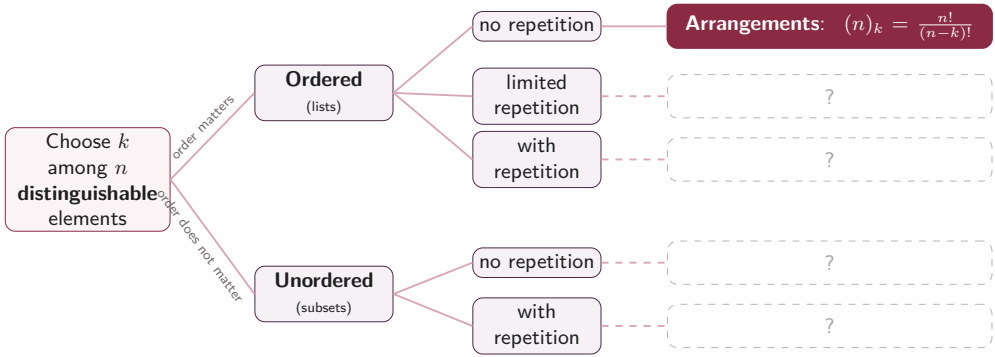
Where are we? Arrangements

First leaf: **ordered** selections **without repetition**, the arrangements.



Where are we? Arrangements

First leaf: **ordered** selections **without repetition**, the arrangements.



k-permutations without repetition

Definition

A **k-permutation without repetition** is an arrangement of k **distinct** elements of S: an ordered sequence of k different elements.

Example

From the previous examples, 123, 789 and 258 are 3-permutations **without** repetition from the first set, while AAB and CDC are 3-permutations **with** repetition from the second set.

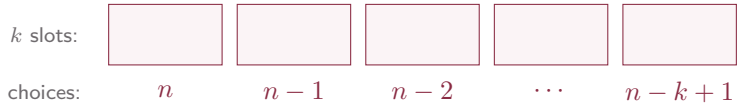
Counting k-permutations without repetition

Theorem (Number of k-permutations without repetition)

For $0 \leq k \leq n$, the number of k-permutations without repetition from a set of n elements, denoted $(n)_k$ (also written A_n^k), is

$$(n)_k = n \times (n-1) \times (n-2) \times \cdots \times (n-k+1).$$

✎ **Proof on the board**



Exercise

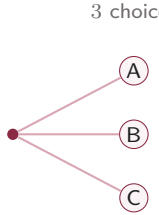
Exercise

- Let $S = \{A, B, C\}$. How many 2-permutations without repetition can be formed? List them.
- Let $S = \{A, B, C, D, E, F\}$. How many 4-permutations without repetition can be formed?

Solution

Solution

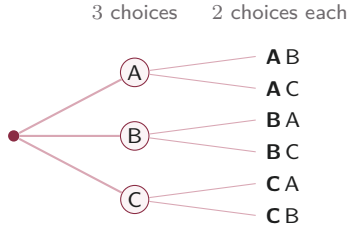
- $(3)_2 = 3 \times 2 = 6$: $AB,\ AC,\ BA,\ BC,\ CA,\ CB.$
- $(6)_4 = 6 \times 5 \times 4 \times 3 = 360.$



Solution

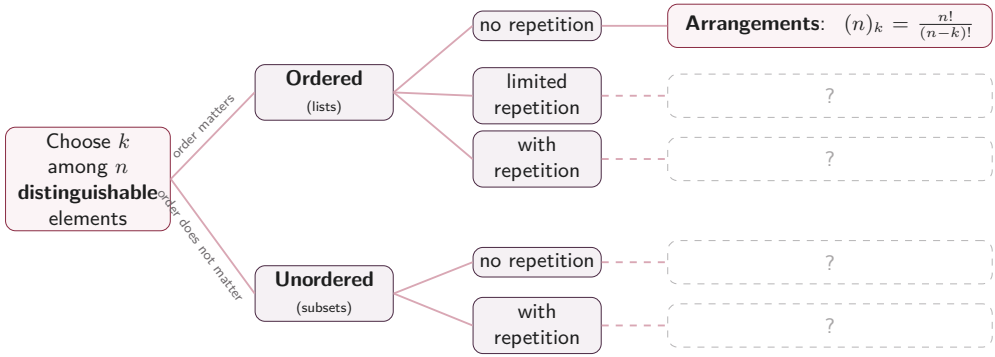
Solution

- $(3)_2 = 3 \times 2 = 6$: $AB,\ AC,\ BA,\ BC,\ CA,\ CB.$
- $(6)_4 = 6 \times 5 \times 4 \times 3 = 360.$



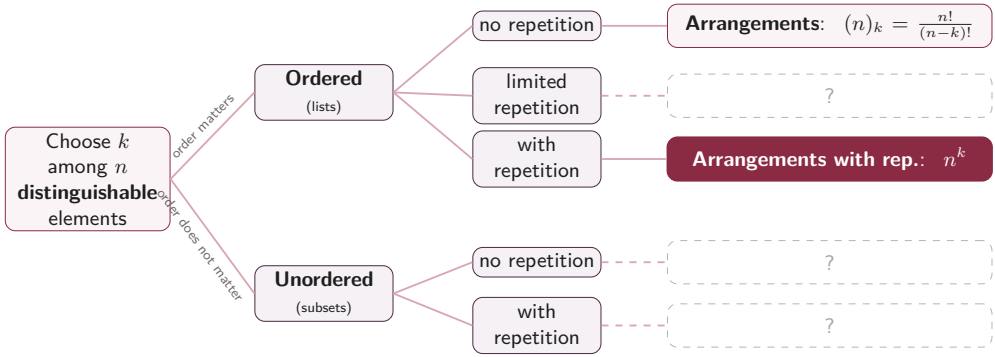
Where are we? Arrangements with repetition

Second leaf: **ordered** selections **with repetition**.



Where are we? Arrangements with repetition

Second leaf: **ordered** selections **with repetition**.



k-permutations with repetition

Definition

A **k-permutation with repetition** is an arrangement of k elements of S where repetition is allowed: each element can appear more than once.

Example

112 and 344 are 3-permutations with repetition from the first set, while AAB and CDC are 3-permutations with repetition from the second set.

Counting k-permutations with repetition

Theorem (Number of k-permutations with repetition)

The number of k-permutations with repetition from a set of n elements, denoted $\mathcal{P}(n, k)$, is

$$\mathcal{P}(n, k) = n^k.$$

✎ **Proof on the board**



The repetition here is allowed and unlimited. Careful: some textbooks (e.g. Rosen) use $P(n, k)$ for $(n)_k$, the count *without* repetition.

Exercise

✎ Exercise

Let $S = \{1, 2, 3\}$. How many 2-permutations with repetition can be formed? List them.

Solution

Solution

Here $n = 3$ and $k = 2$, so $\mathcal{P}(3, 2) = 3^2 = 9$:

$$11,\ 12,\ 13,\ 21,\ 22,\ 23,\ 31,\ 32,\ 33.$$

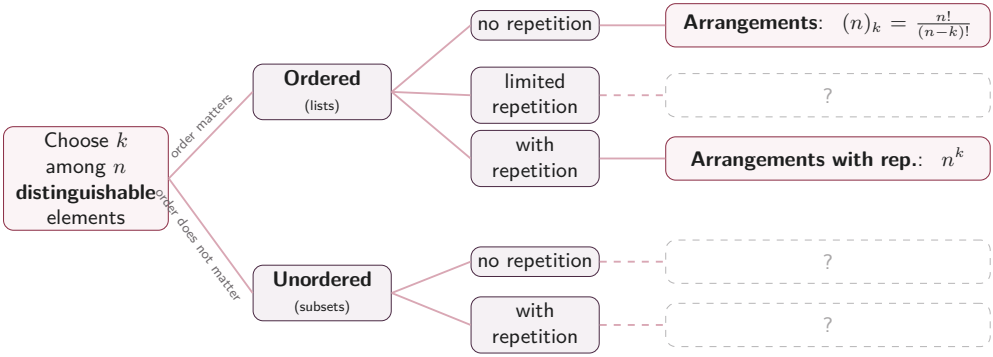
This is exactly the $\{H, I, M\}$ exercise of the introduction.

1st \ 2nd	**1**	**2**	**3**
1	11	12	13
2	21	22	23
3	31	32	33

$3 \times 3 = 9$

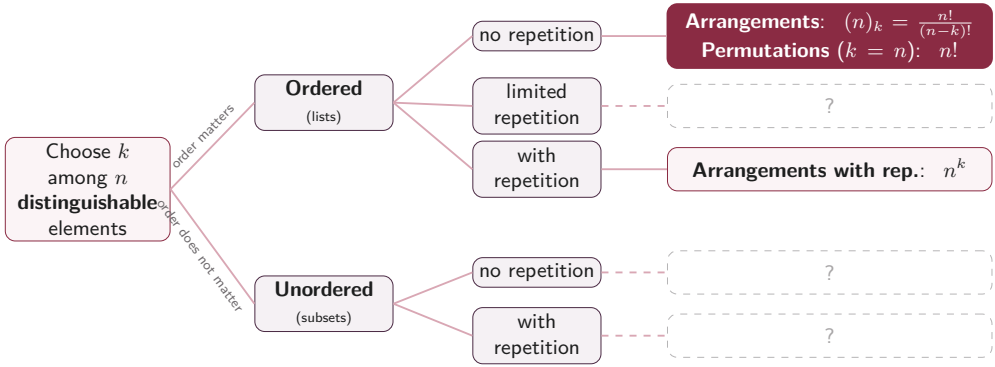
Where are we? Permutations

Special case of the first leaf: arrange **all** the n elements ($k = n$).



Where are we? Permutations

Special case of the first leaf: arrange **all** the n elements ($k = n$).



Permutations (without repetition)

For a non-negative integer n, the **factorial** of n is

$$n! = n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1, \qquad 0! = 1$$

($0!$ is the empty product; it keeps the rule $n! = n\,(n-1)!$ valid for $n = 1$.)

Definition

A **permutation** of $S = \{e_1, \ldots, e_n\}$ is an arrangement of **all** the elements of S without repetition (an n-permutation without repetition).

Example

$S = \{A, B, C\}$: $ABC,\ ACB,\ BAC,\ BCA,\ CAB,\ CBA,$ so 6 permutations.

Number of permutations

Theorem (Number of Permutations)

The number of permutations of a set with n elements, denoted P_n, is

$$P_n = n!.$$

✎ **Proof on the board**

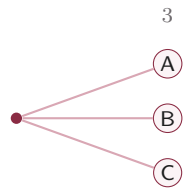
Number of permutations

Theorem (Number of Permutations)

The number of permutations of a set with n elements, denoted P_n, is

$$P_n = n!.$$

✎ **Proof on the board**



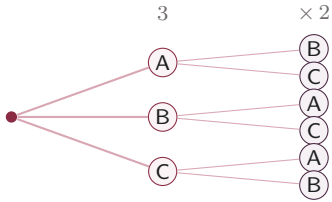
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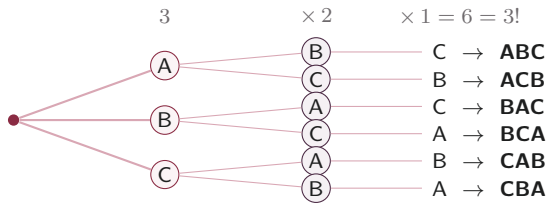
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✎ **Proof on the board**



Exercise

✎ Exercise

How many nonsense words of length 26 can be formed using all the letters of the English alphabet without repetition?

Solution

Solution

This is a permutation of the 26 letters: $P_{26} = 26! \approx 4.03 \times 10^{26}$.
Compare with the $26^{26} \approx 6.16 \times 10^{36}$ words of the introduction, where repetition was allowed.

Relations between the counts

Proposition

- For all $n \geq 0$: $\quad P_n = (n)_n = n!$.
- For $0 \leq k \leq n$: $\quad (n)_k = \dfrac{n!}{(n-k)!}$.
- $P_n = (n)_k \, P_{n-k}$.

✎ Exercise

Prove the proposition.

Solution: Relations between the counts

Solution

- A permutation of S is an n-permutation without repetition, so $P_n = (n)_n = n(n-1)\cdots 1 = n!$.
- For $0 \le k \le n$, split the product $n!$ after its first k factors:

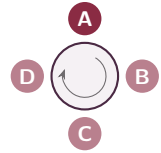
$$n! = \underbrace{n(n-1)\cdots(n-k+1)}_{(n)_k} \cdot \underbrace{(n-k)(n-k-1)\cdots 1}_{(n-k)!}, \qquad \text{so} \qquad (n)_k = \frac{n!}{(n-k)!}$$

  (for $k = n$ the second product is empty and equals $0! = 1$).
- This is the same identity: $P_n = n! = (n)_k\,(n-k)! = (n)_k\,P_{n-k}$.
  Combinatorially: to arrange the n elements, fill the first k positions ($(n)_k$ ways), then arrange the $n-k$ remaining elements in the last $n-k$ positions (P_{n-k} ways), and use the multiplication principle.

Circular permutations

Definition (Circular permutation)

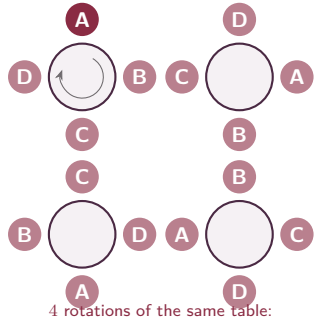
A **circular permutation** of n distinct elements is a way to place them **around a circle** (for example, people around a round table).
Two placements are the **same** if one is a **rotation** of the other: everybody moves by the same number of seats, so everybody keeps the **same neighbours**.



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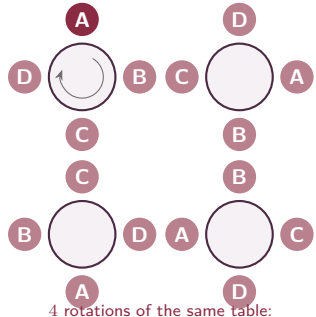
4 rotations of the same table:
one circular permutation

Circular permutations

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Two placements are the **same** if one is a **rotation** of the other: everybody moves by the same number of seats, so everybody keeps the **same neighbours**.

In a **line**, $ABCD$, $DABC$, $CDAB$, $BCDA$ are 4 **different** permutations. Around the table, they are the **same** circular permutation: we do not count the rotations.



4 rotations of the same table:
one circular permutation

Number of circular permutations

Theorem (Number of circular permutations)

For $n \geq 1$, the number of circular permutations of n distinct elements is $\frac{n!}{n} = (n-1)!$.

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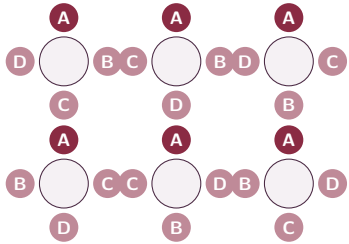
- **Way 1:** each circular permutation gives exactly n linear permutations (its n rotations). So we divide $n!$ by n.

Number of circular permutations

Theorem (Number of circular permutations)

For $n \geq 1$, the number of circular permutations of n distinct elements is $\dfrac{n!}{n} = (n-1)!$.

- **Way 1:** each circular permutation gives exactly n linear permutations (its n rotations). So we divide $n!$ by n.
- **Way 2:** the rotations do not matter, so we can **fix** the seat of A. Then we place the $n-1$ others in the $n-1$ free seats, as in a line: $(n-1)!$.



A fixed at the top:
$3! = 6$ circular permutations of A, B, C, D

Exercise

✎ Exercise

8 people sit around a round table (rotations are not counted).

- In how many ways can they sit?
- In how many ways can they sit if two friends A and B want to sit next to each other?

Solution

Solution

- Circular permutations of 8 people: $(8-1)! = 7! = 5040$.
- Glue A and B together as one block. We now have 7 objects around the table: $(7-1)! = 6! = 720$ ways. Inside the block, A and B can be in 2 orders (AB or BA).
  Total: $2 \times 6! = 1440$.

See also Worksheet 1, Exercises 10 and 11 (round table).

3.2

Combinations, Binomial Theorem & Pascal's Triangle

Unordered selections: subsets

Combinations

Definition

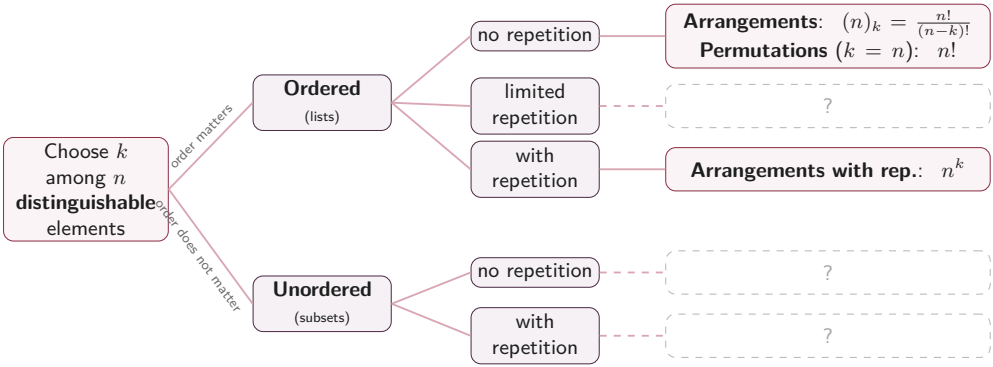
A **combination** is a selection of a subset from a finite set, where the order of selection does not matter. Simply put, a combination of a set is a **subset** of that set.

Examples

- $\{1, 2, 4\}$ and $\{5, 4, 1\}$ are combinations of size 3 from $\{1, 2, 3, 4, 5\}$.
- $\{A, B, C\}$ and $\{B, A, C\}$ are the **same** combination from $\{A, B, C, D\}$: the order does not matter.

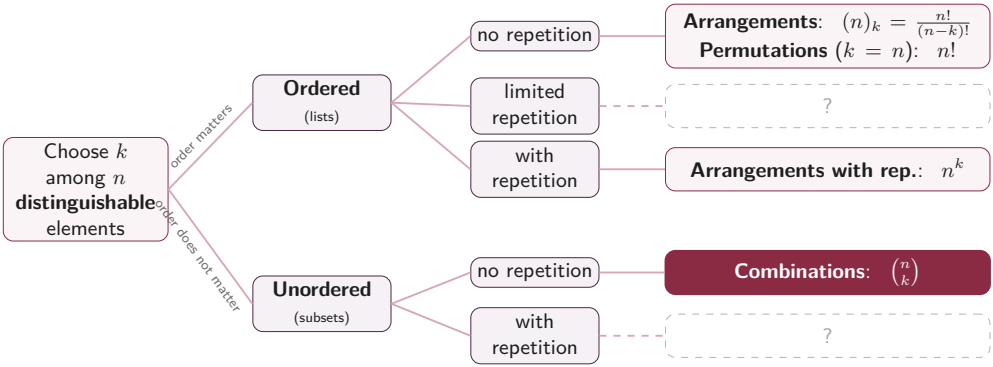
Where are we? Combinations

We now forget the order: **unordered** selections **without repetition**, the combinations.



Where are we? Combinations

We now forget the order: **unordered** selections **without repetition**, the combinations.



k-combinations without repetition

Definition

A **k-combination without repetition** from a set S of n elements is a selection of k elements of S where order does not matter: a **subset** of k elements.

Examples

- The 2-combinations of $\{A, B, C, D\}$:
  $\{A, B\}, \{A, C\}, \{A, D\}, \{B, C\}, \{B, D\}, \{C, D\}$, so 6.
- All the combinations of $S = \{1, 2, 3\}$:
  - 0: $\{\}$ 1: $\{1\}, \{2\}, \{3\}$ 2: $\{1, 2\}, \{1, 3\}, \{2, 3\}$ 3: $\{1, 2, 3\}$

  so 8 combinations in total: all the subsets of S, including the empty set.

Number of k-combinations without repetition

Theorem (Number of k-combinations without repetition)

For $0 \leq k \leq n$, the number of k-combinations without repetition from a set of n elements, denoted $\binom{n}{k}$ (also written C_n^k), is

$$\binom{n}{k} = \frac{(n)_k}{k!} = \frac{n!}{k!\,(n-k)!}.$$

The numbers $\binom{n}{k}$ are called **binomial coefficients**.

✎ **Proof on the board: double counting**

Why do we divide by $k!$?

$\binom{4}{3} = 4$ **subsets**

$\{A, B, C\}$

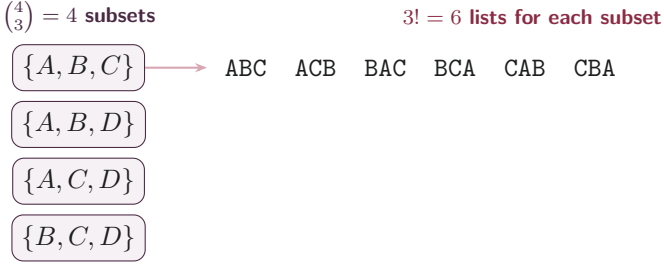
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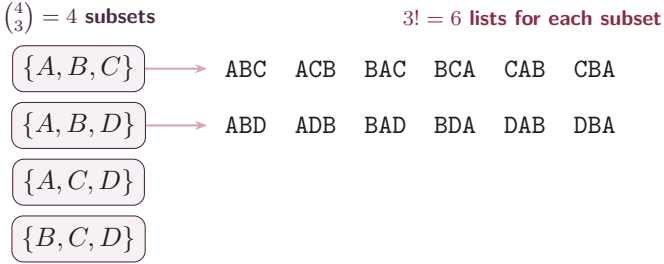
$\{B, C, D\}$

$3! = 6$ **lists for each subset**

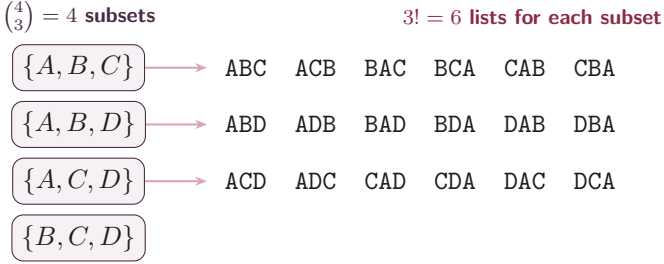
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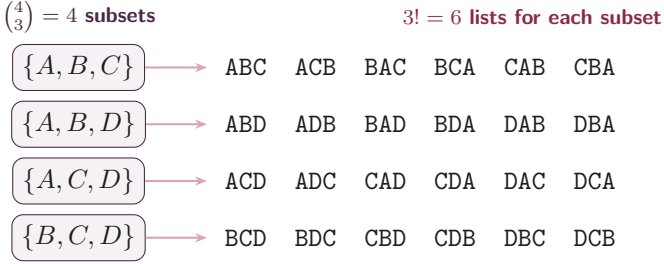
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$$(4)_3 = 24 = 3! \times \binom{4}{3} = 6 \times 4, \quad \text{so} \quad \binom{n}{k} = \frac{(n)_k}{k!}$$

Exercise

✎ Exercise

- How many ways can we select a team of 3 people from a group of 5 persons $S = \{P1, P2, P3, P4, P5\}$? List all the possible teams.
- From a group of 10 people (6 boys and 4 girls), how many ways can we select a team containing
  - 4 people?
  - 2 boys and 2 girls?

Solution

Solution

- $\binom{5}{3} = \frac{5!}{3!\,2!} = \frac{5 \times 4}{2!} = 10$ teams:

  $\{P1, P2, P3\}, \{P1, P2, P4\}, \{P1, P2, P5\}, \{P1, P3, P4\}, \{P1, P3, P5\},$

  $\{P1, P4, P5\}, \{P2, P3, P4\}, \{P2, P3, P5\}, \{P2, P4, P5\}, \{P3, P4, P5\}.$

- 4 people: $\binom{10}{4} = \frac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1} = 210.$
- 2 boys and 2 girls: $\binom{6}{2} \times \binom{4}{2} = 15 \times 6 = 90.$

Pascal's triangle

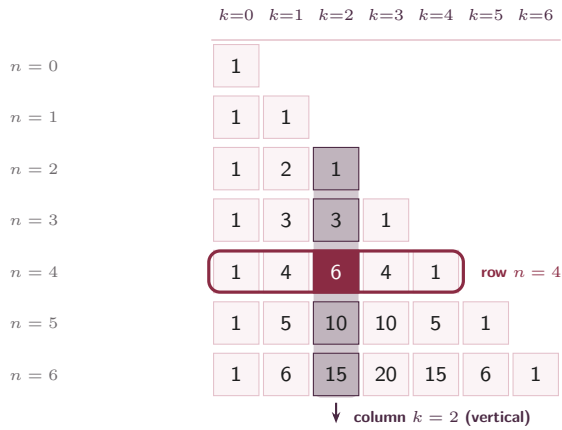
Arrange the binomial coefficients $\binom{n}{k}$ with $n =$ row number and $k =$ position in the row (column number): this is the **binomial coefficients triangle**, or **Pascal's triangle**.

$$\binom{n}{0} = 1 \quad \text{and} \quad \binom{n}{n} = 1.$$

$n \backslash k$	0	1	2	3	4	5	6	7	8	9
0	1									
1	1	1								
2	1	2	1							
3	1	3	3	1						
4	1	4	6	4	1					
5	1	5	10	10	5	1				
6	1	6	15	20	15	6	1			
7	1	7	21	35	35	21	7	1		
8	1	8	28	56	70	56	28	8	1	
9	1	9	36	84	126	126	84	36	9	1

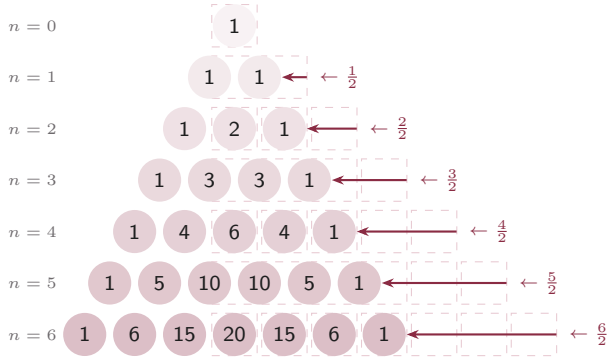
From the table to the triangle

- **Table:** row n is **horizontal**, column k is **vertical**.



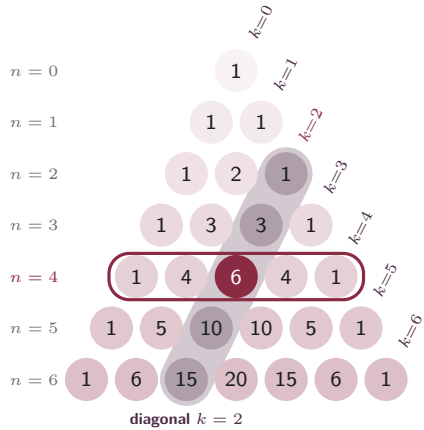
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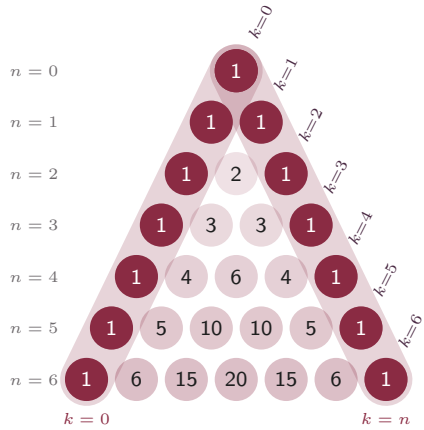


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- **Triangle:** row n stays horizontal, column k becomes a **diagonal** $\swarrow$, with k written at its top.
- $k = 0$ gives the **left border**, $k = n$ gives the **right border**: all equal to 1.

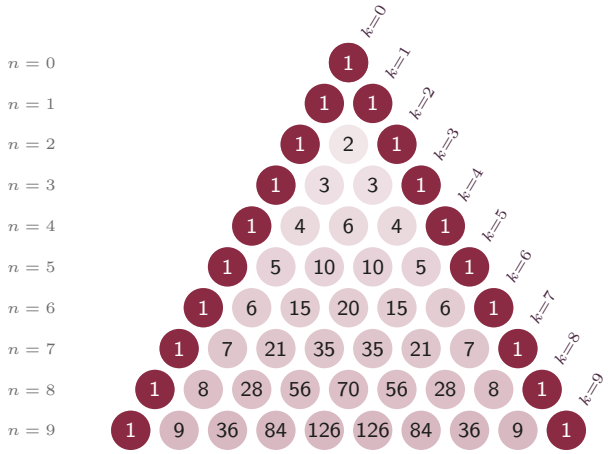
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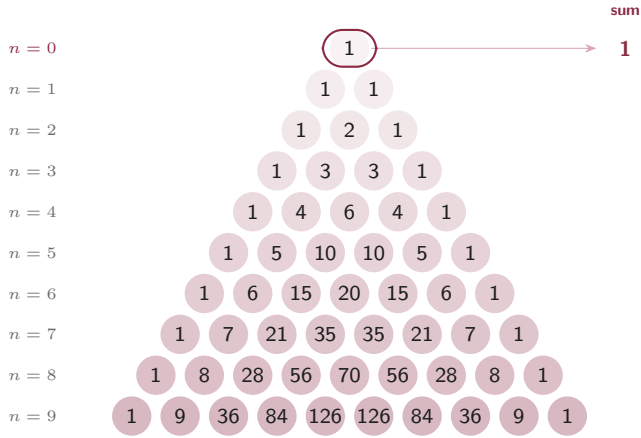


Pascal's triangle

Rows $\rightarrow$ powers of 2:

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

row $n = 0$ $(k = 0)$:
$1 = 1 = 2^0$

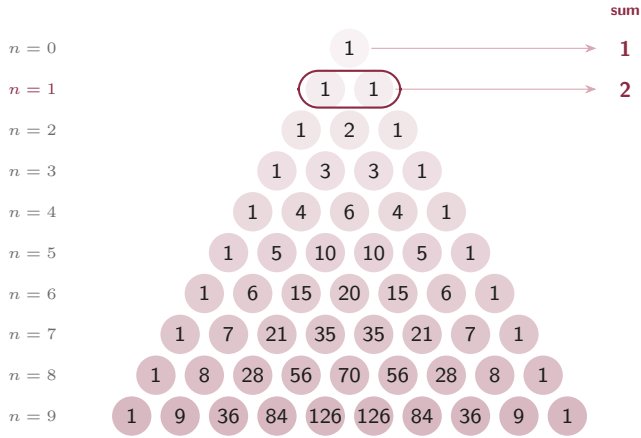


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row $n = 1$ $(k = 0, \ldots, 1)$:
$1 + 1 = 2 = 2^1$

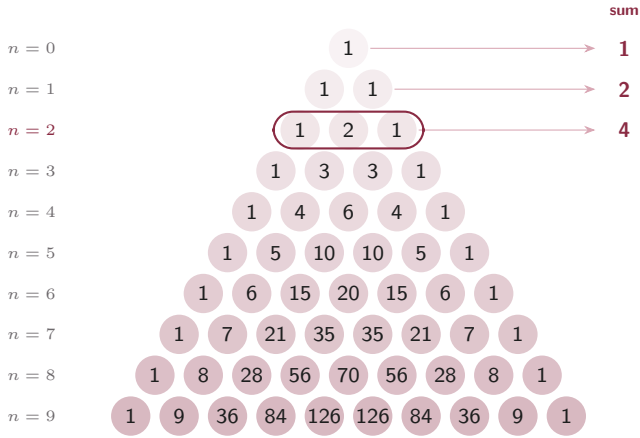


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row $n = 2$ $(k = 0, \ldots, 2)$:
$1 + 2 + 1 = 4 = 2^2$

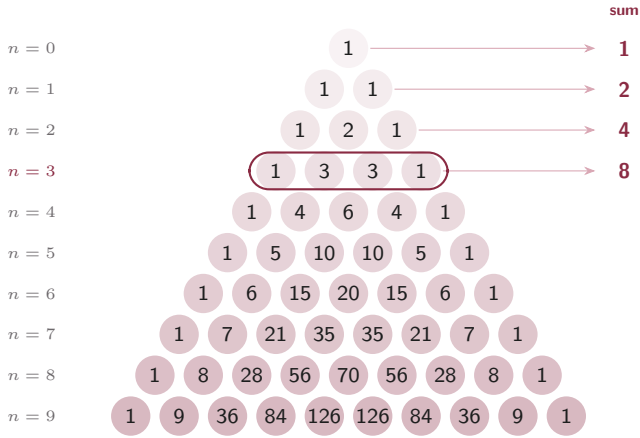


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row $n = 3$ $(k = 0, \ldots, 3)$:
$1 + 3 + 3 + 1 = 8 = 2^3$

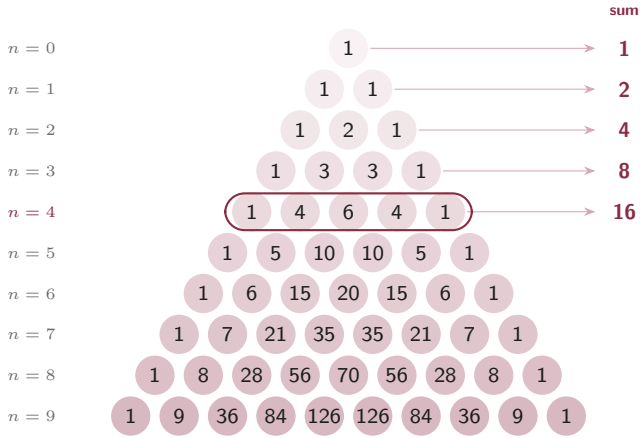


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row $n = 4$ $(k = 0, \ldots, 4)$:
$1 + 4 + 6 + 4 + 1 = 16 = 2^4$

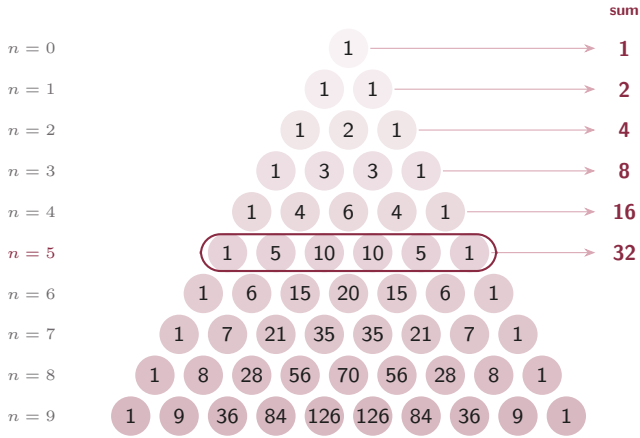


Pascal's triangle

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row $n = 5$ $(k = 0, \ldots, 5)$:
$1+5+10+10+5+1 = 32 = 2^5$

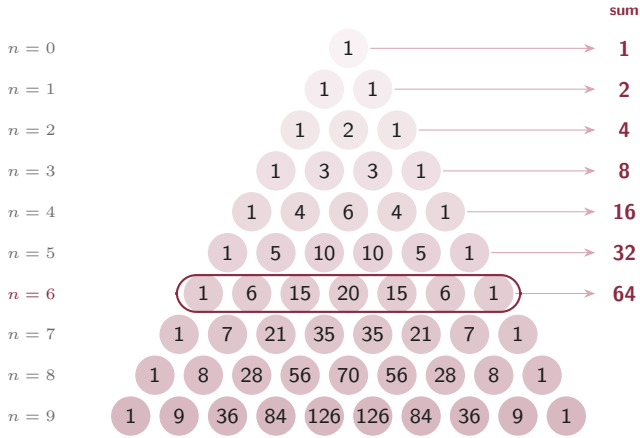


Pascal's triangle

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row $n = 6$ $(k = 0, \ldots, 6)$:
$1 + 6 + 15 + 20 + 15 + 6 + 1 = 64 = 2^6$

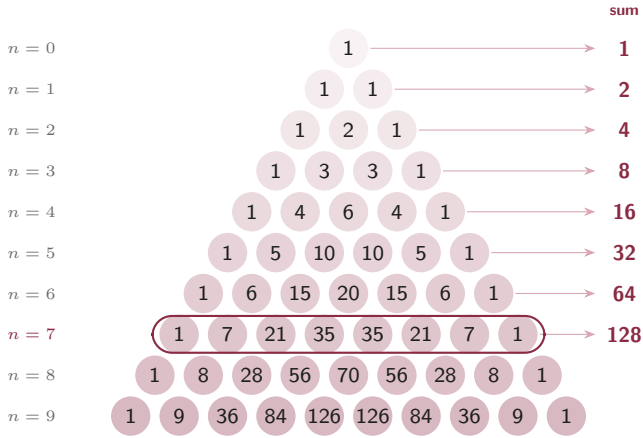


Pascal's triangle

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row $n = 7$ $(k = 0, \ldots, 7)$:
$1+7+21+35+35+21+7+1 = 128 = 2^7$

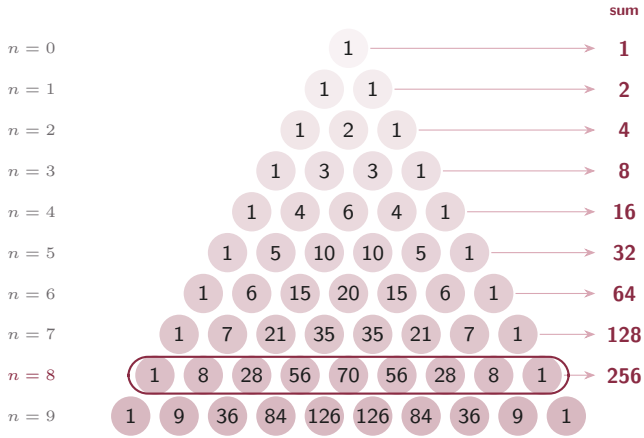


Pascal's triangle

Rows $\rightarrow$ powers of 2:

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row $n = 8$ $(k = 0, \ldots, 8)$:
$1 + 8 + 28 + 56 + 70 + 56 + 28 + 8 + 1 = 256 = 2^8$

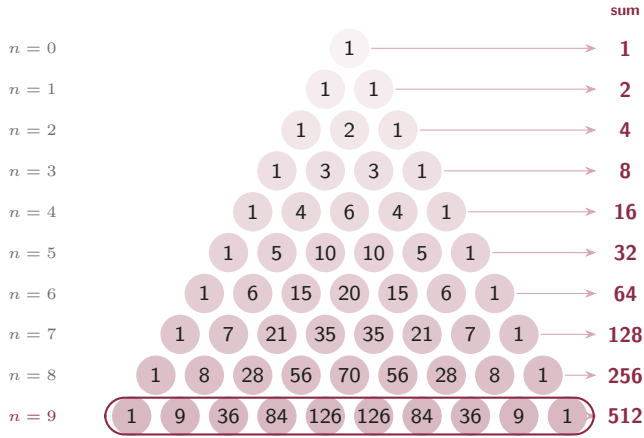


Pascal's triangle

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row $n = 9$ $(k = 0, \ldots, 9)$:
$1 + 9 + 36 + 84 + 126 + 126 + 84 + 36 + 9 + 1 = 512 = 2^9$



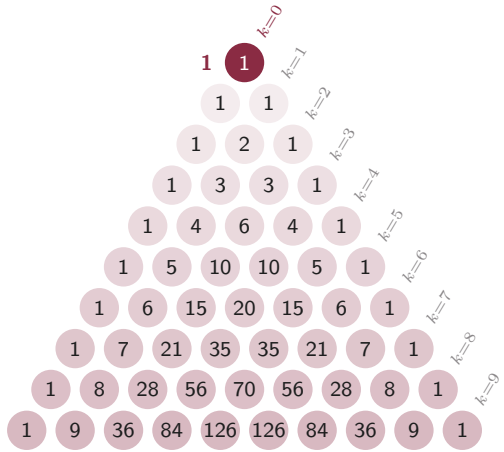
Pascal's triangle

Shallow diagonals ↗
Fibonacci numbers:

$$\sum_{j\geq 0} \binom{m-j}{j} = F_{m+1}$$

term $\binom{m-j}{j}$: row $n = m - j$,
diagonal $k = j$.

diagonal $m = 0$:
$\binom{0}{0}$
$= 1 = 1$



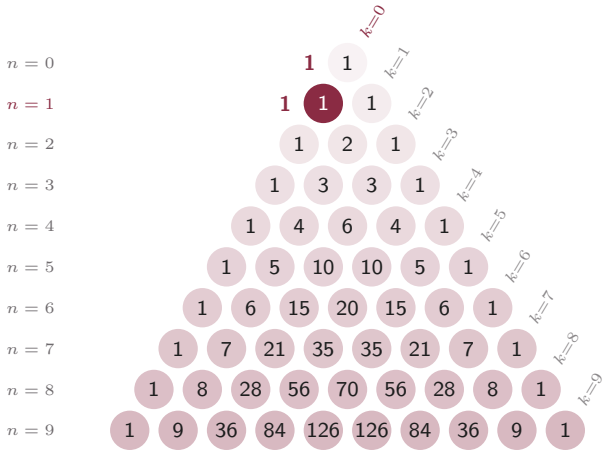
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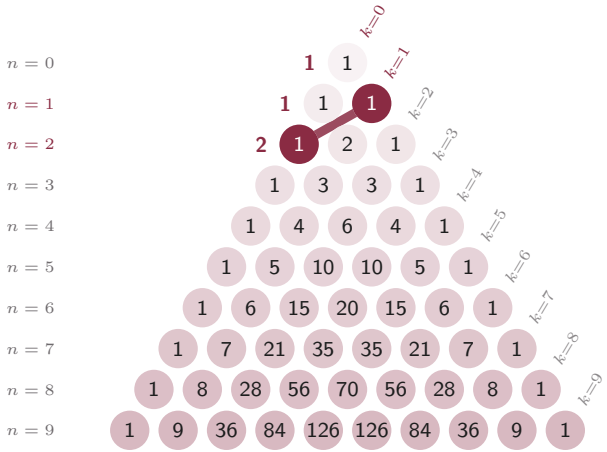
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Shallow diagonals ↗
Fibonacci numbers:

$$\sum_{j\geq 0} \binom{m-j}{j} = F_{m+1}$$

term $\binom{m-j}{j}$: row $n = m - j$,
diagonal $k = j$.

diagonal $m = 2$:
$\binom{2}{0} + \binom{1}{1}$
$= 1 + 1 = 2$



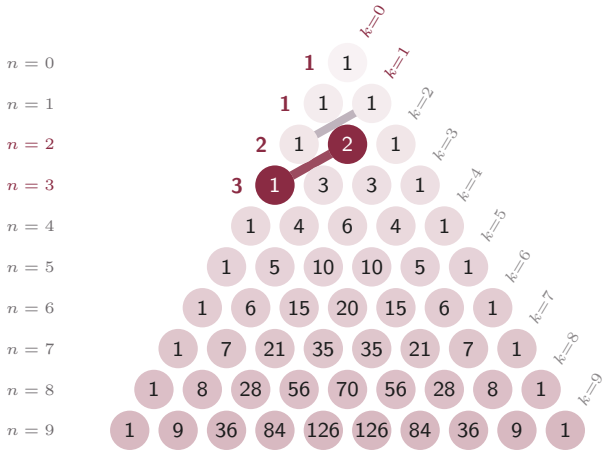
Pascal's triangle

Shallow diagonals ↗
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$$\sum_{j\geq 0} \binom{m-j}{j} = F_{m+1}$$

term $\binom{m-j}{j}$: row $n = m - j$,
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diagonal $m = 3$:
$\binom{3}{0} + \binom{2}{1}$
$= 1 + 2 = 3$



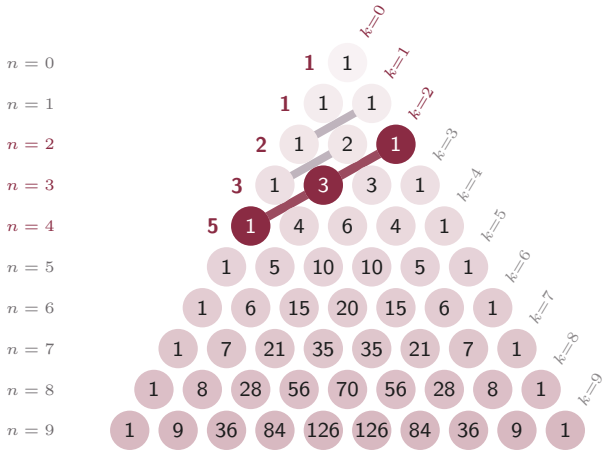
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Shallow diagonals ↗
Fibonacci numbers:

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term $\binom{m-j}{j}$: row $n = m - j$,
diagonal $k = j$.

diagonal $m = 4$:
$\binom{4}{0} + \binom{3}{1} + \binom{2}{2}$
$= 1 + 3 + 1 = 5$



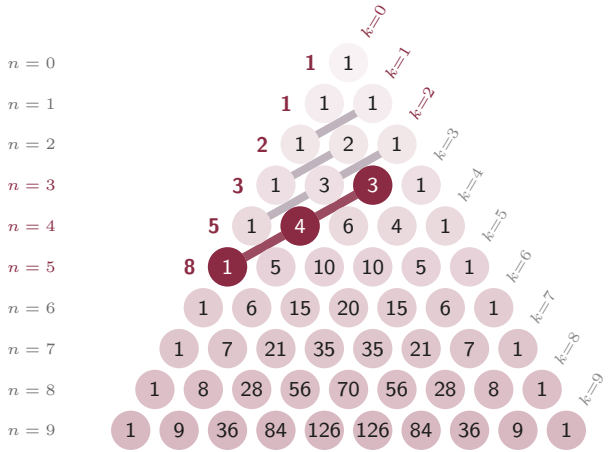
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term $\binom{m-j}{j}$: row $n = m - j$,
diagonal $k = j$.

diagonal $m = 5$:
$\binom{5}{0} + \binom{4}{1} + \binom{3}{2}$
$= 1 + 4 + 3 = 8$



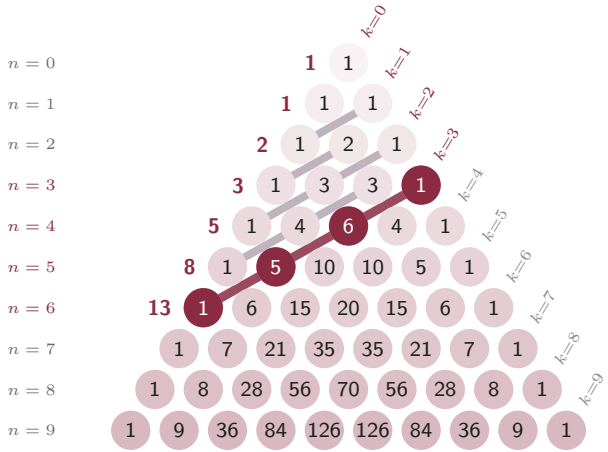
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term $\binom{m-j}{j}$: row $n = m - j$,
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diagonal $m = 6$:
$\binom{6}{0} + \binom{5}{1} + \binom{4}{2} + \binom{3}{3}$
$= 1 + 5 + 6 + 1 = 13$



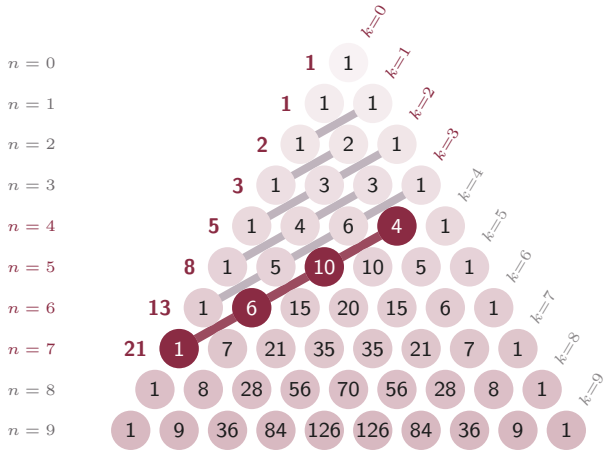
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diagonal $m = 7$:
$\binom{7}{0} + \binom{6}{1} + \binom{5}{2} + \binom{4}{3}$
$= 1 + 6 + 10 + 4 = 21$



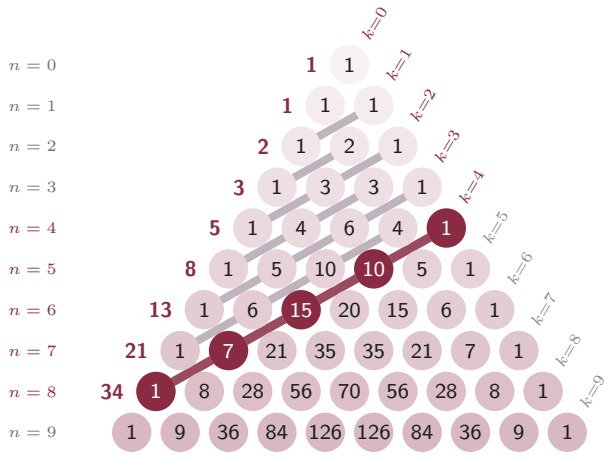
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term $\binom{m-j}{j}$: row $n = m - j$,
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diagonal $m = 8$:
$\binom{8}{0} + \binom{7}{1} + \binom{6}{2} + \binom{5}{3} + \binom{4}{4}$
$= 1 + 7 + 15 + 10 + 1 = 34$



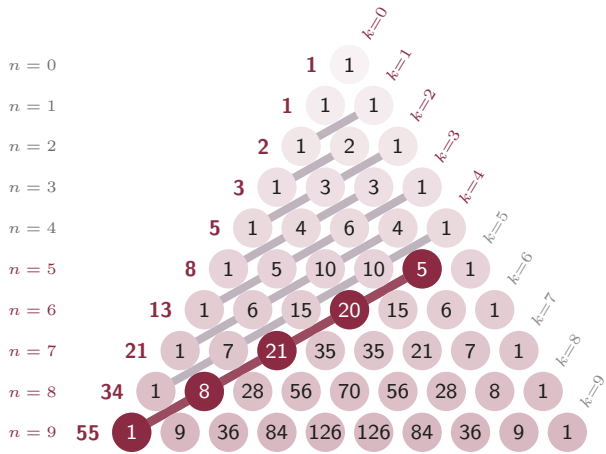
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term $\binom{m-j}{j}$: row $n = m - j$,
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diagonal $m = 9$:
$\binom{9}{0} + \binom{8}{1} + \binom{7}{2} + \binom{6}{3} + \binom{5}{4}$
$= 1 + 8 + 21 + 20 + 5 = 55$



Symmetry

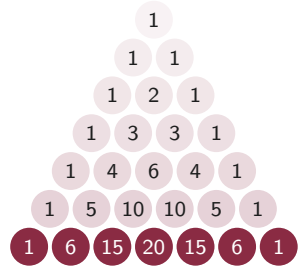
Theorem (Symmetry Property)

For $0 \leq k \leq n$: $$\binom{n}{k} = \binom{n}{n-k}.$$

✎ **Algebraic and combinatorial proofs on the board**

In Pascal's triangle, every row reads the same from left to right and from right to left:

1 6 15 20 15 6 1



Symmetry

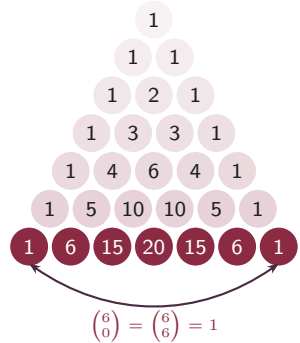
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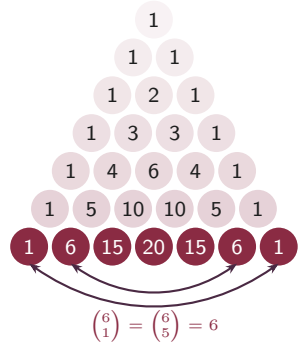
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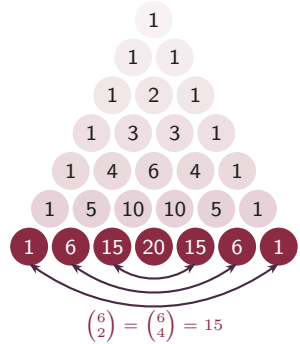
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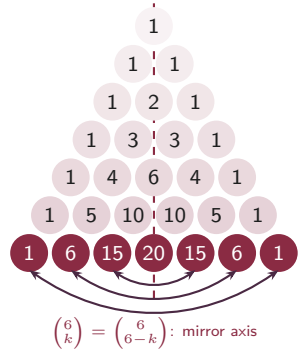
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✎ **Algebraic and combinatorial proofs on the board**

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Pascal's rule

Theorem (Recurrence Relation (Pascal's rule))

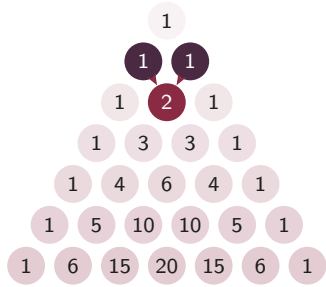
For $n \geq 1$ and $1 \leq k \leq n-1$:

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

✎ **Proof on the board**

Remark

For now, if $k > n$, $k < 0$ or $n < 0$, we set $\binom{n}{k} = 0$.



Pascal's rule

Theorem (Recurrence Relation (Pascal's rule))

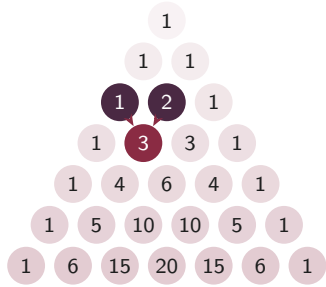
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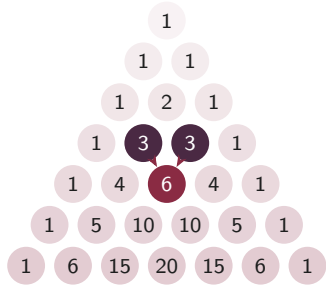
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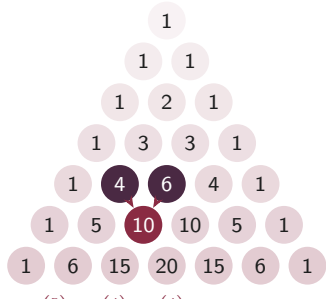
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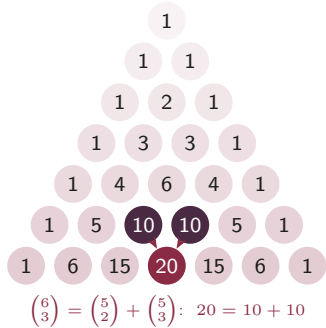
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$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

✎ **Proof on the board**

Remark

For now, if $k > n$, $k < 0$ or $n < 0$, we set $\binom{n}{k} = 0$.



Vandermonde's identity

Theorem (Vandermonde's Identity)

For any non-negative integers m, n and k:

$$\binom{m+n}{k} = \sum_{r=0}^{k} \binom{m}{r}\binom{n}{k-r}.$$

✎ **Proof on the board: double counting**

Worksheet 1 – Exercise 20 (lecture)

✎ **Exercise (Worksheet 1, Exercise 20)**

Prove in two different ways the following identity (known as the absorption/extraction identity), for $1 \leq k \leq n$:

$$k\binom{n}{k} = n\binom{n-1}{k-1}.$$

Worksheet 1 – Exercise 20 (lecture)

✎ Exercise (Worksheet 1, Exercise 20)

Prove in two different ways the following identity (known as the absorption/extraction identity), for $1 \leq k \leq n$:

$$k\binom{n}{k} = n\binom{n-1}{k-1}.$$

Two ways

- **Way 1 (algebra):** use the formula $\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$.
- **Way 2 (counting):** find one question whose answer is the left side *and* the right side.

Exercise 20: first way (with the formula)

Solution (Exercise 20 – algebraic proof)

We start from the left side and use $\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$:

$$k\binom{n}{k} = k \cdot \frac{n!}{k!\,(n-k)!} \qquad \text{(definition of } \tbinom{n}{k}\text{)}$$

Exercise 20: first way (with the formula)

Solution (Exercise 20 – algebraic proof)

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$$k\binom{n}{k} = k \cdot \frac{n!}{k!\,(n-k)!} \qquad \text{(definition of } \tbinom{n}{k}\text{)}$$

$$= \frac{n!}{(k-1)!\,(n-k)!} \qquad \text{because } \frac{k}{k!} = \frac{1}{(k-1)!}$$

Exercise 20: first way (with the formula)

Solution (Exercise 20 – algebraic proof)

We start from the left side and use $\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$:

$$
\begin{aligned}
k\binom{n}{k} &= k \cdot \frac{n!}{k!\,(n-k)!} && \text{(definition of } \tbinom{n}{k}\text{)} \\
&= \frac{n!}{(k-1)!\,(n-k)!} && \text{because } \tfrac{k}{k!} = \tfrac{1}{(k-1)!} \\
&= n \cdot \frac{(n-1)!}{(k-1)!\,(n-k)!} && \text{because } n! = n \cdot (n-1)!
\end{aligned}
$$

Exercise 20: first way (with the formula)

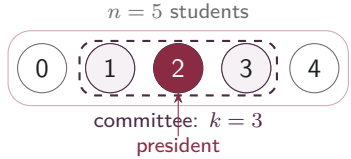
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&= \frac{n!}{(k-1)!\,(n-k)!} && \text{because } \tfrac{k}{k!} = \tfrac{1}{(k-1)!} \\
&= n \cdot \frac{(n-1)!}{(k-1)!\,(n-k)!} && \text{because } n! = n \cdot (n-1)! \\
&= n\binom{n-1}{k-1} && \text{because } (n-1) - (k-1) = n-k.
\end{aligned}
$$

Exercise 20: second way (by counting)

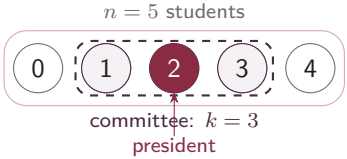
Question: in a group of n students, we choose a **committee** of k students, and **one** of them is the **president**. How many ways?



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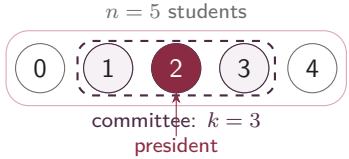
- **Way 1:** choose the committee: $\binom{n}{k}$; then the president among its k members: k.
  Total: $k\binom{n}{k}$.



Exercise 20: second way (by counting)

Question: in a group of n students, we choose a **committee** of k students, and **one** of them is the **president**. How many ways?

- **Way 1:** choose the committee: $\binom{n}{k}$; then the president among its k members: k.
  Total: $k\binom{n}{k}$.
- **Way 2:** choose the president first: n; then the $k-1$ other members among the $n-1$ other students: $\binom{n-1}{k-1}$.
  Total: $n\binom{n-1}{k-1}$.

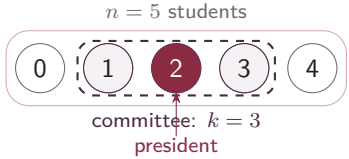


Exercise 20: second way (by counting)

Question: in a group of n students, we choose a **committee** of k students, and **one** of them is the **president**. How many ways?

- **Way 1:** choose the committee: $\binom{n}{k}$; then the president among its k members: k.
  Total: $k\binom{n}{k}$.
- **Way 2:** choose the president first: n; then the $k-1$ other members among the $n-1$ other students: $\binom{n-1}{k-1}$.
  Total: $n\binom{n-1}{k-1}$.

We counted the same thing twice, so the two numbers are **equal**.



check: $3 \cdot \binom{5}{3} = 3 \cdot 10 = 30$

$5 \cdot \binom{4}{2} = 5 \cdot 6 = 30$

Worksheet 1 – Exercise 21 (lecture)

✎ Exercise (Worksheet 1, Exercise 21)

Establish the "committee-subcommittee" identity, for $0 \leq r \leq k \leq n$:

$$\binom{n}{k}\binom{k}{r} = \binom{n}{r}\binom{n-r}{k-r}.$$

Worksheet 1 – Exercise 21 (lecture)

✎ Exercise (Worksheet 1, Exercise 21)

Establish the "committee-subcommittee" identity, for $0 \leq r \leq k \leq n$:

$$\binom{n}{k}\binom{k}{r} = \binom{n}{r}\binom{n-r}{k-r}.$$

Idea

The name of the identity gives the question: choose a **committee**, then a **subcommittee** inside it.

Exercise 21: first way (by counting)

Question: from n students, we choose a **committee** of k students, and inside it a **subcommittee** of r students. How many ways?



Exercise 21: first way (by counting)

Question: from n students, we choose a **committee** of k students, and inside it a **subcommittee** of r students. How many ways?

- **Way 1:** choose the committee: $\binom{n}{k}$; then the subcommittee inside it: $\binom{k}{r}$.
  Total: $\binom{n}{k}\binom{k}{r}$.



Exercise 21: first way (by counting)

Question: from n students, we choose a **committee** of k students, and inside it a **subcommittee** of r students. How many ways?

- **Way 1:** choose the committee: $\binom{n}{k}$; then the subcommittee inside it: $\binom{k}{r}$.
  Total: $\binom{n}{k}\binom{k}{r}$.
- **Way 2:** choose the subcommittee first: $\binom{n}{r}$; then the $k-r$ other committee members among the $n-r$ remaining students: $\binom{n-r}{k-r}$.
  Total: $\binom{n}{r}\binom{n-r}{k-r}$.



Exercise 21: first way (by counting)

Question: from n students, we choose a **committee** of k students, and inside it a **subcommittee** of r students. How many ways?

- **Way 1:** choose the committee: $\binom{n}{k}$; then the subcommittee inside it: $\binom{k}{r}$.
  Total: $\binom{n}{k}\binom{k}{r}$.
- **Way 2:** choose the subcommittee first: $\binom{n}{r}$; then the $k-r$ other committee members among the $n-r$ remaining students: $\binom{n-r}{k-r}$.
  Total: $\binom{n}{r}\binom{n-r}{k-r}$.

Same question, so the two numbers are **equal**.



check ($n = 5$, $k = 3$, $r = 2$):

$$\binom{5}{3}\binom{3}{2} = 10 \cdot 3 = 30$$
$$\binom{5}{2}\binom{3}{1} = 10 \cdot 3 = 30$$

Exercise 21: second way (with the formula)

Solution (Exercise 21 – algebraic proof)

Left side:

$$\binom{n}{k}\binom{k}{r} = \frac{n!}{k!\,(n-k)!} \cdot \frac{k!}{r!\,(k-r)!}$$

Exercise 21: second way (with the formula)

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$$\binom{n}{k}\binom{k}{r} = \frac{n!}{k!\,(n-k)!} \cdot \frac{k!}{r!\,(k-r)!} = \frac{n!}{r!\,(k-r)!\,(n-k)!}$$

($k!$ is on top and at the bottom: it cancels.)

Exercise 21: second way (with the formula)

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Left side:

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Right side:

$$\binom{n}{r}\binom{n-r}{k-r} = \frac{n!}{r!\,(n-r)!} \cdot \frac{(n-r)!}{(k-r)!\,(n-k)!} = \frac{n!}{r!\,(k-r)!\,(n-k)!}$$

($(n-r)!$ cancels, and $(n-r)-(k-r) = n-k$.)

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($(n-r)!$ cancels, and $(n-r)-(k-r) = n-k$.)

The two sides give the **same** fraction, so the identity is true.

The Binomial Theorem

Theorem (The Binomial Theorem)

For any real numbers a, b and every non-negative integer n:

$$(a+b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \cdots + \binom{n}{n}b^n = \sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k.$$

✎ **Combinatorial proof on the board; induction as exercise**

Corollary

For every $n \geq 1$: $\displaystyle\sum_{k=0}^{n}\binom{n}{k} = 2^n$ and $\displaystyle\sum_{k=0}^{n}(-1)^k\binom{n}{k} = 0.$

The binomial theorem: a picture

$$(a+b)^3 = \boxed{(a+b)}\boxed{(a+b)}\boxed{(a+b)}$$

The binomial theorem: a picture

$$(a+b)^3 = \boxed{(a+b)}\boxed{(a+b)}\boxed{(a+b)}$$

choose a or b in each factor: $2^3 = 8$ words

aaa	aab aba baa	abb bab bba	bbb

The binomial theorem: a picture

$$(a+b)^3 = \boxed{(a+b)}\boxed{(a+b)}\boxed{(a+b)}$$

choose a or b in each factor: $2^3 = 8$ words

aaa	aab aba baa	abb bab bba	bbb
0 b: $\binom{3}{0} = 1$	1 b: $\binom{3}{1} = 3$	2 b: $\binom{3}{2} = 3$	3 b: $\binom{3}{3} = 1$

The binomial theorem: a picture

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aaa	aab aba baa	abb bab bba	bbb
0 b: $\binom{3}{0} = 1$	1 b: $\binom{3}{1} = 3$	2 b: $\binom{3}{2} = 3$	3 b: $\binom{3}{3} = 1$

$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ (1 3 3 1 = row 3 of Pascal's triangle)

Solution: The Binomial Theorem by induction

Solution (proof by induction on n)

For $n = 0$ both sides equal 1. Assume the formula for n. Then

$$(a+b)^{n+1} = (a+b)\sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k = \sum_{k=0}^{n}\binom{n}{k}a^{n+1-k}b^k + \sum_{k=1}^{n+1}\binom{n}{k-1}a^{n+1-k}b^k,$$

(we put $k \to k-1$ in the second sum). With the convention $\binom{n}{-1} = \binom{n}{n+1} = 0$ and Pascal's rule $\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}$:

$$(a+b)^{n+1} = \sum_{k=0}^{n+1}\binom{n+1}{k}a^{n+1-k}b^k.$$

Corollary: take $a = b = 1$, then $a = 1$, $b = -1$ (for $n \geq 1$).

Exercise

✎ Exercise

Determine the coefficient of $a^{17}b^9$ in the expansion of $(a+b)^{26}$.

Solution

Solution

The general term is $\binom{26}{k}a^{26-k}b^k$. We need $26-k=17$, so $k=9$. The coefficient is

$$\binom{26}{9} = 3\,124\,550.$$

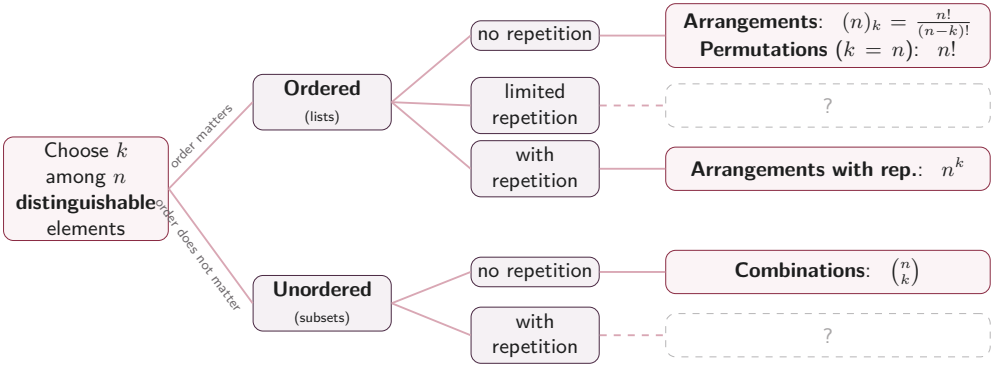
3.3

Permutations with Limited Repetition & Multinomials

The MISSISSIPPI problem

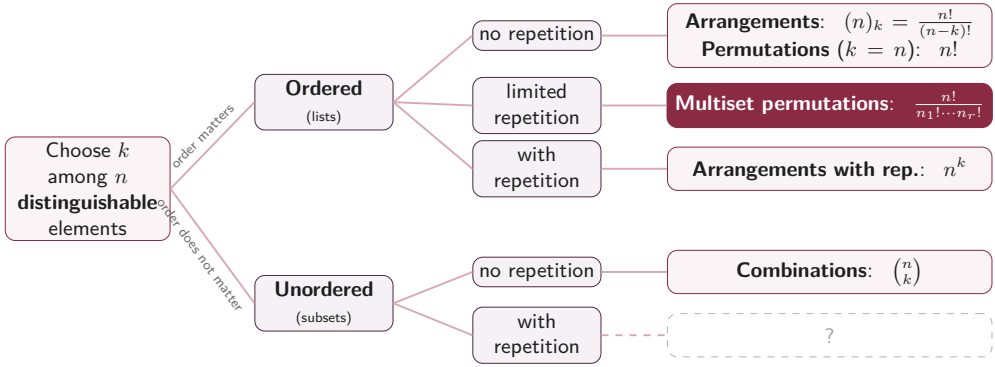
Where are we? Permutations of a multiset

Ordered again, each element used a **limited** number of times (MISSISSIPPI).



Where are we? Permutations of a multiset

Ordered again, each element used a **limited** number of times (MISSISSIPPI).



The MISSISSIPPI problem



How many distinct words can be formed by rearranging the letters of "MISSISSIPPI"?

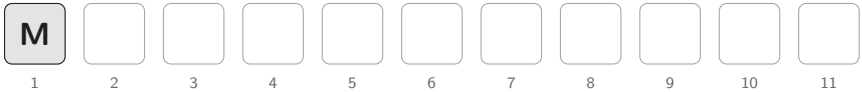
- Not $11!$: permuting identical letters gives the **same** word.
- Choose the **positions** of each letter instead:

$$\underbrace{\binom{11}{1}}_{M}\underbrace{\binom{10}{4}}_{I}\underbrace{\binom{6}{4}}_{S}\underbrace{\binom{2}{2}}_{P} = \frac{11!}{1!\,10!}\cdot\frac{10!}{4!\,6!}\cdot\frac{6!}{4!\,2!}\cdot\frac{2!}{2!} = \frac{11!}{1!\,4!\,4!\,2!} = 34\,650.$$

MISSISSIPPI: choosing the positions

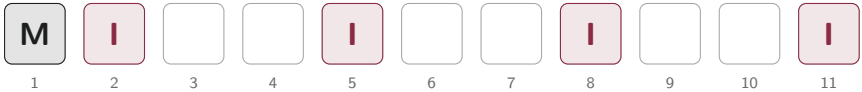


MISSISSIPPI: choosing the positions



positions of the M's: choose 1 among the 11 free positions: $\binom{11}{1} = 11$

MISSISSIPPI: choosing the positions



positions of the I's: choose 4 among the 10 free positions: $\binom{10}{4} = 210$

MISSISSIPPI: choosing the positions



positions of the S's: choose 4 among the 6 free positions: $\binom{6}{4} = 15$

MISSISSIPPI: choosing the positions



positions of the P's: choose 2 among the 2 free positions: $\binom{2}{2} = 1$

MISSISSIPPI: choosing the positions



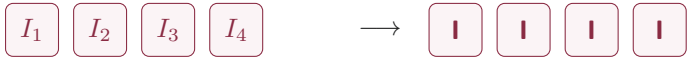
$$\binom{11}{1}\binom{10}{4}\binom{6}{4}\binom{2}{2} = 11 \times 210 \times 15 \times 1 = 34\,650$$

Another way to see it

Double counting

Label the letters $I_1, \ldots, I_4$, $S_1, \ldots, S_4$, P_1, P_2. There are $11!$ arrangements of these 11 **distinct** letters, and each word is obtained exactly $4!\,4!\,2!$ times (by permuting the labels). Hence there are

$$\frac{11!}{4!\,4!\,2!} \text{ words.}$$



$4!$ labelled versions give the same word

Permutations with limited repetition

Theorem (Permutations with Limited Repetition)

Let S be a collection of k distinct elements $e_1, \ldots, e_k$, where e_i is repeated n_i times, with $n_1 + n_2 + \cdots + n_k = n$. The number of permutations of all the elements of S is

$$\binom{n}{n_1, n_2, \ldots, n_k} = \binom{n}{n_1}\binom{n-n_1}{n_2}\cdots\binom{n-\sum_{i=1}^{k-1} n_i}{n_k} = \frac{n!}{n_1!\, n_2! \cdots n_k!}.$$

The numbers $\binom{n}{n_1,\ldots,n_k}$ are called **multinomial coefficients**.

✎ **Proof on the board**

Multisets

Remarks

- The collection S is denoted $\{n_1 \cdot e_1,\ n_2 \cdot e_2,\ \ldots,\ n_k \cdot e_k\}$ and is called a **multiset**: unlike a set, it can contain the same element several times.
- Infinite repetition is denoted by ∞: in $\{\infty \cdot a,\ 5 \cdot b,\ 3 \cdot c\}$, the element a appears infinitely often, b five times and c three times.
- Problems with infinite repetition are generally simpler than those with finite repetition. For $A = \{\infty \cdot a_1, \ldots, \infty \cdot a_n\}$, the number of arrangements of length k is

$$n^k,$$

which are exactly the k-permutations with repetition.

Properties of multinomial coefficients

Multinomial coefficients count the ways to divide n distinct objects into k **labelled** (distinguishable) groups of sizes $n_1, \ldots, n_k$, with $n_1 + \cdots + n_k = n$.

Theorem (Recursive Formula for Multinomial Coefficients)

Let $n \geq 1$, $k \geq 2$, and $n_1, \ldots, n_k \geq 0$ with $n_1 + \cdots + n_k = n$. Then

$$\binom{n}{n_1, \ldots, n_k} = \sum_{i=1}^{k} \binom{n-1}{n_1, \ldots, n_i - 1, \ldots, n_k}$$

(a term with a negative entry $n_i - 1 < 0$ is taken to be 0; for $k = 2$ this is Pascal's rule).

✎ **Exercise.** Prove the theorem both algebraically and combinatorially.

Solution: Recursive formula for multinomial coefficients

Solution

Combinatorial proof. $\binom{n}{n_1,\ldots,n_k}$ counts the words of length n containing n_i letters e_i. Classify these words by their *first* letter. If the first letter is e_i (possible only if $n_i \geq 1$), the remaining $n-1$ letters form a word with n_i-1 letters e_i and n_j letters e_j for $j \neq i$: there are $\binom{n-1}{n_1,\ldots,n_i-1,\ldots,n_k}$ of them. The cases are disjoint; add them.

Algebraic proof. For $n_i \geq 1$,

$$\binom{n-1}{n_1,\ldots,n_i-1,\ldots,n_k} = \frac{(n-1)!}{n_1!\cdots(n_i-1)!\cdots n_k!} = \frac{(n-1)!}{n_1!\cdots n_k!}\, n_i,$$

and this is also true (both sides are 0) when $n_i = 0$. Summing over i:

$$\sum_{i=1}^{k} \binom{n-1}{n_1,\ldots,n_i-1,\ldots,n_k} = \frac{(n-1)!}{n_1!\cdots n_k!}\,(n_1+\cdots+n_k) = \frac{n!}{n_1!\cdots n_k!}.$$

The Multinomial Theorem

Theorem (Multinomial Theorem)

For any positive integer n and any real numbers $x_1, \ldots, x_k$:

$$(x_1 + x_2 + \cdots + x_k)^n = \sum_{\substack{n_1+\cdots+n_k=n \\ n_1,\ldots,n_k \geq 0}} \binom{n}{n_1, n_2, \ldots, n_k} x_1^{n_1} x_2^{n_2} \cdots x_k^{n_k}.$$

✎ **Exercise**

- Give a combinatorial proof, using arguments similar to the binomial theorem.
- Prove the multinomial theorem by induction on n, and by induction on k.

Solution: Multinomial Theorem, combinatorial proof

Solution (combinatorial proof)

Write $(x_1 + x_2 + \cdots + x_k)^n$ as a product of n identical factors. Expanding, each term corresponds to choosing one of $x_1, \ldots, x_k$ in each of the n factors, that is, to a word of length n over the letters $x_1, \ldots, x_k$.
The term $x_1^{n_1} x_2^{n_2} \cdots x_k^{n_k}$ is obtained from the words containing exactly n_1 letters x_1, …, n_k letters x_k. By the previous theorem, there are

$$\binom{n}{n_1, n_2, \ldots, n_k}$$

such words, which is the coefficient of $x_1^{n_1} \cdots x_k^{n_k}$.

Solution: Multinomial Theorem by induction on n

Solution (proof by induction on n)

For $n = 1$ the formula is $x_1 + \cdots + x_k$. Assume it for n and multiply by $(x_1 + \cdots + x_k)$. A monomial $x_1^{n_1} \cdots x_k^{n_k}$ with $n_1 + \cdots + n_k = n + 1$ is obtained from $x_i \cdot x_1^{n_1} \cdots x_i^{n_i - 1} \cdots x_k^{n_k}$, for $i = 1, \ldots, k$. Its coefficient is therefore

$$\sum_{i=1}^{k} \binom{n}{n_1, \ldots, n_i - 1, \ldots, n_k} = \binom{n+1}{n_1, \ldots, n_k}$$

by the recursive formula for multinomial coefficients.

Solution: Multinomial Theorem by induction on k

Solution (proof by induction on k)

For $k = 2$ this is the binomial theorem. Assume the result for $k - 1$ variables. By the binomial theorem,

$$(x_1 + \cdots + x_k)^n = \sum_{n_k=0}^{n} \binom{n}{n_k} (x_1 + \cdots + x_{k-1})^{n-n_k} \, x_k^{n_k},$$

and by the induction hypothesis the coefficient of $x_1^{n_1} \cdots x_k^{n_k}$ is

$$\binom{n}{n_k} \binom{n - n_k}{n_1, \ldots, n_{k-1}} = \frac{n!}{n_k! \, (n - n_k)!} \cdot \frac{(n - n_k)!}{n_1! \cdots n_{k-1}!} = \binom{n}{n_1, \ldots, n_k}.$$

Example: $(x_1 + x_2 + x_3)^3$

$$
\begin{aligned}
(x_1 + x_2 + x_3)^3 &= \sum_{\substack{n_1+n_2+n_3=3 \\ n_1,n_2,n_3 \geq 0}} \binom{3}{n_1, n_2, n_3} x_1^{n_1} x_2^{n_2} x_3^{n_3} \\
&= \binom{3}{0,0,3} x_3^3 + \binom{3}{0,3,0} x_2^3 + \binom{3}{0,1,2} x_2 x_3^2 + \binom{3}{0,2,1} x_2^2 x_3 + \binom{3}{1,1,1} x_1 x_2 x_3 \\
&\quad + \binom{3}{1,2,0} x_1 x_2^2 + \binom{3}{1,0,2} x_1 x_3^2 + \binom{3}{2,0,1} x_1^2 x_3 + \binom{3}{2,1,0} x_1^2 x_2 + \binom{3}{3,0,0} x_1^3
\end{aligned}
$$

Example: $(x_1 + x_2 + x_3)^3$, computing the coefficients

Computing the coefficients

$$x_1^3 + x_2^3 + x_3^3 + 3\left(x_1^2x_2 + x_1^2x_3 + x_1x_2^2 + x_2^2x_3 + x_1x_3^2 + x_2x_3^2\right) + 6\,x_1x_2x_3.$$

There are 10 terms: we will see that $10 = \left(\!\binom{3}{3}\!\right) = \binom{5}{3}$.

Exercise

✎ Exercise

Without expanding completely, find the coefficients in $(a+b+2c)^5$ of the following terms (watch the factor 2):

- a^2b^2c
- a^3bc
- b^5

Solution

Solution

The general term of $(a+b+2c)^5$ is

$$\binom{5}{n_1, n_2, n_3} a^{n_1} b^{n_2} (2c)^{n_3} = \binom{5}{n_1, n_2, n_3} 2^{n_3} a^{n_1} b^{n_2} c^{n_3}.$$

- a^2b^2c: $\dfrac{5!}{2!\,2!\,1!} \times 2 = 30 \times 2 = 60.$
- a^3bc: $\dfrac{5!}{3!\,1!\,1!} \times 2 = 20 \times 2 = 40.$
- b^5: $\dfrac{5!}{0!\,5!\,0!} \times 2^0 = 1.$

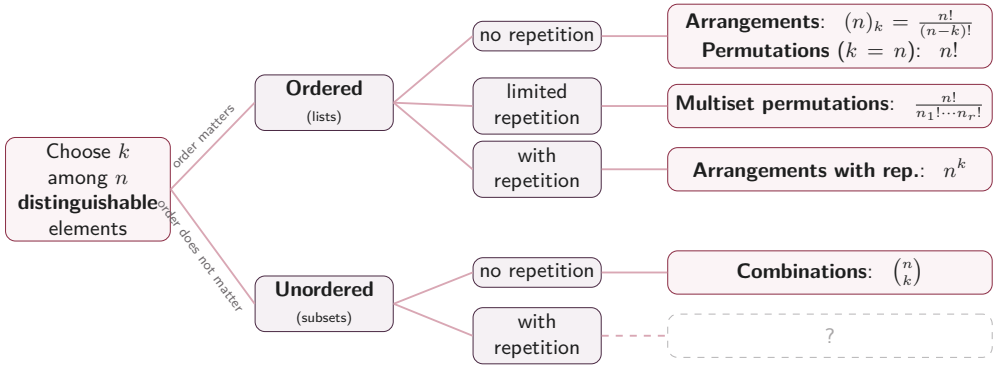
3.4

Combinations with Repetition

Unordered selections where repetition is allowed

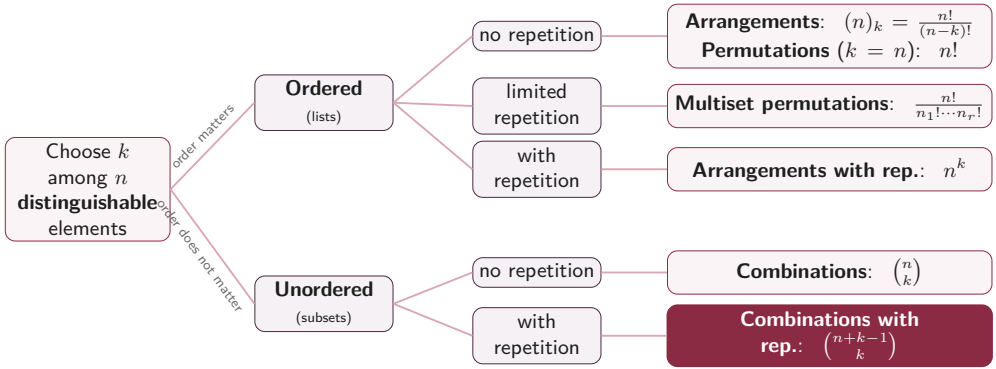
Where are we? Combinations with repetition

Last leaf: **unordered** selections **with repetition**.



Where are we? Combinations with repetition

Last leaf: **unordered** selections **with repetition**.



Combinations with repetition

Let $S = \{1, 2, 3\}$. The 2-combinations of S:

- **without** repetition: $\{1, 2\},\ \{1, 3\},\ \{2, 3\}$;
- **with** repetition: $\{1, 1\},\ \{1, 2\},\ \{1, 3\},\ \{2, 2\},\ \{2, 3\},\ \{3, 3\}$, so 6 solutions.

If $S = \{1, 2, \ldots, n\}$, how many k-combinations with repetition are there?

Definition

A k-**combination with repetition** from $S = \{1, 2, \ldots, n\}$ is an **unordered** selection of k elements, where repetition of elements is allowed.

Combinations with repetition and multisets

Remarks

- If $x_1, \ldots, x_n$ are the repetitions of $e_1, \ldots, e_n$, then

$$x_1 + x_2 + \cdots + x_n = k, \qquad 0 \leq x_i \leq k.$$

- A k-combination with repetition is the same as a **multiset** of k elements $\{x_1 \cdot e_1, \ldots, x_n \cdot e_n\}$ with $x_1 + \cdots + x_n = k$.

Number of k-combinations with repetition

Theorem (Number of k-combinations with repetition)

The number of k-combinations with repetition from a set of n elements, denoted $\binom{n}{k}$, is

$$\left(\!\binom{n}{k}\!\right) = \binom{n+k-1}{k} = \binom{n+k-1}{n-1}.$$

✎ **Proof on the board: stars and bars**

Stars and bars

$$\underbrace{\star\,\star\,\cdots\,\star}_{x_1 \text{ stars}} \;\Big|\; \underbrace{\star\,\star\,\cdots\,\star}_{x_2 \text{ stars}} \;\Big|\; \cdots \;\Big|\; \underbrace{\star\,\star\,\cdots\,\star}_{x_n \text{ stars}} \qquad k \text{ stars, } n-1 \text{ bars}$$

Back to $S = \{1, 2, 3\}$ and $k = 2$ (3 groups, so 2 bars)

$$\{1, 1\} \quad \longleftrightarrow \quad \star\star \,||$$

Stars and bars

$$\underbrace{\star\ \star\ \cdots\ \star}_{x_1 \text{ stars}} \Big| \underbrace{\star\ \star\ \cdots\ \star}_{x_2 \text{ stars}} \Big| \cdots \Big| \underbrace{\star\ \star\ \cdots\ \star}_{x_n \text{ stars}}$$

k stars, $n-1$ bars

Back to $S = \{1, 2, 3\}$ and $k = 2$ (3 groups, so 2 bars)

$$\{1, 1\} \longleftrightarrow \star\star\,||$$
$$\{1, 2\} \longleftrightarrow \star|\star\,|$$

Stars and bars

$$\underbrace{\star\,\star\,\cdots\star}_{x_1 \text{ stars}} \,\Big|\, \underbrace{\star\,\star\,\cdots\star}_{x_2 \text{ stars}} \,\Big|\, \cdots \,\Big|\, \underbrace{\star\,\star\,\cdots\star}_{x_n \text{ stars}}$$

k stars, $n-1$ bars

Back to $S = \{1, 2, 3\}$ and $k = 2$ (3 groups, so 2 bars)

$$\{1,1\} \longleftrightarrow \star\star\,||$$
$$\{1,2\} \longleftrightarrow \star|\star\,|$$
$$\{1,3\} \longleftrightarrow \star||\star$$

Stars and bars

$$\underbrace{\star\,\star\,\cdots\star}_{x_1 \text{ stars}} \,\Big|\, \underbrace{\star\,\star\,\cdots\star}_{x_2 \text{ stars}} \,\Big|\, \cdots \,\Big|\, \underbrace{\star\,\star\,\cdots\star}_{x_n \text{ stars}} \qquad k \text{ stars, } n-1 \text{ bars}$$

Back to $S = \{1, 2, 3\}$ and $k = 2$ (3 groups, so 2 bars)

$$\begin{aligned}
\{1,1\} &\longleftrightarrow \star\star\,||\\
\{1,2\} &\longleftrightarrow \star|\star\,|\\
\{1,3\} &\longleftrightarrow \star||\star\\
\{2,2\} &\longleftrightarrow |\,\star\star|
\end{aligned}$$

Stars and bars

$$\underbrace{\star\,\star\,\cdots\,\star}_{x_1 \text{ stars}} \,\Big|\, \underbrace{\star\,\star\,\cdots\,\star}_{x_2 \text{ stars}} \,\Big|\, \cdots \,\Big|\, \underbrace{\star\,\star\,\cdots\,\star}_{x_n \text{ stars}} \qquad k \text{ stars, } n-1 \text{ bars}$$

Back to $S = \{1, 2, 3\}$ and $k = 2$ (3 groups, so 2 bars)

$$\begin{aligned}
\{1,1\} &\longleftrightarrow \star\star\,||\\
\{1,2\} &\longleftrightarrow \star|\star\,|\\
\{1,3\} &\longleftrightarrow \star||\star\\
\{2,2\} &\longleftrightarrow |\,\star\star|\\
\{2,3\} &\longleftrightarrow |\,\star\,|\star
\end{aligned}$$

Stars and bars

$$\underbrace{\star\,\star\,\cdots\star}_{x_1 \text{ stars}} \,\Big|\, \underbrace{\star\,\star\,\cdots\star}_{x_2 \text{ stars}} \,\Big|\, \cdots \,\Big|\, \underbrace{\star\,\star\,\cdots\star}_{x_n \text{ stars}} \qquad k \text{ stars, } n-1 \text{ bars}$$

Back to $S = \{1, 2, 3\}$ and $k = 2$ (3 groups, so 2 bars)

$$\begin{aligned}
\{1,1\} &\longleftrightarrow \star\star\,||\\
\{1,2\} &\longleftrightarrow \star|\,\star\,|\\
\{1,3\} &\longleftrightarrow \star||\star\\
\{2,2\} &\longleftrightarrow |\,\star\star|\\
\{2,3\} &\longleftrightarrow |\,\star\,|\star\\
\{3,3\} &\longleftrightarrow ||\,\star\star
\end{aligned}$$

Hence $\left(\!\binom{3}{2}\!\right) = \binom{4}{2} = 6$, as found at the beginning: choose the 2 positions of the stars among 4.

Application: integer solutions of linear equations

Question: how many non-negative integer solutions does the equation

$$x_1 + x_2 + \cdots + x_n = k$$

have?

Theorem

The number of non-negative integer solutions of $x_1 + x_2 + \cdots + x_n = k$ is

$$\left(\!\binom{n}{k}\!\right) = \binom{k+n-1}{k}.$$

Exercises

✎ Exercise

- How many non-negative integer solutions are there for $x_1 + x_2 + x_3 = 6$?
- How many integer solutions exist for $x_1 + x_2 + x_3 + x_4 = 18$, where $x_1 \geq 3$, $x_2 \geq 5$, $x_3 \geq 1$ and $x_4 \geq 0$?

✎ Exercise

How many non-negative integer solutions exist for the inequality

$$x_1 + x_2 + \cdots + x_n \leq k\,?$$

Hint: introduce a slack variable $x_{n+1} \geq 0$.

Solutions

Solution

- $x_1 + x_2 + x_3 = 6$: $\binom{6+3-1}{6} = \binom{8}{2} = 28$.
- Put $y_1 = x_1 - 3$, $y_2 = x_2 - 5$, $y_3 = x_3 - 1$, $y_4 = x_4$: then $y_i \geq 0$ and

$$y_1 + y_2 + y_3 + y_4 = 18 - 9 = 9, \qquad \text{so} \quad \binom{9+4-1}{9} = \binom{12}{3} = 220.$$

- Put $x_{n+1} = k - (x_1 + \cdots + x_n) \geq 0$: the solutions of the inequality correspond exactly to the non-negative solutions of $x_1 + \cdots + x_n + x_{n+1} = k$, so there are

$$\binom{k + (n+1) - 1}{k} = \binom{n+k}{k}.$$

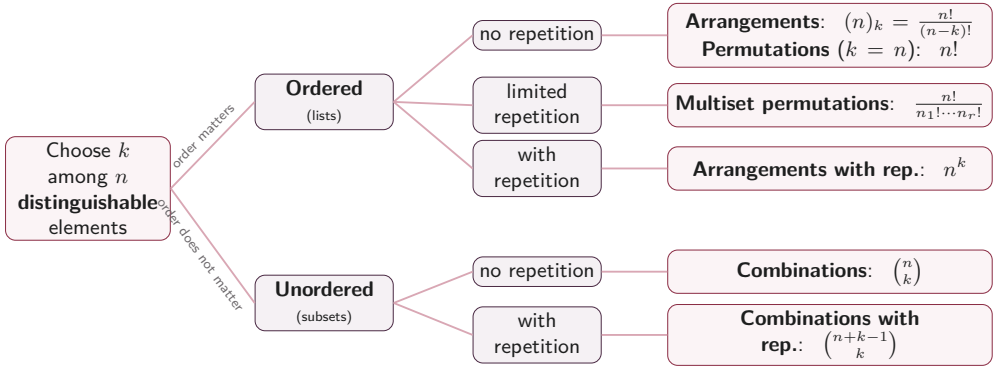
04

Summary

Four ways to select k elements from n

The tree of notions: summary

All the leaves are filled: the whole chapter in one picture.



Selecting k elements from a set of n elements

	Without repetition	**With repetition**
Ordered	$(n)_k = \dfrac{n!}{(n-k)!}$	n^k
Unordered	$\binom{n}{k} = \dfrac{n!}{k!\,(n-k)!}$	$\left(\!\binom{n}{k}\!\right) = \binom{n+k-1}{k}$

Plus: permutations $n!$ and multinomial coefficients $\dfrac{n!}{n_1! \cdots n_k!}$ for multisets.

Back to the opening questions

- Arrangements of "NHSM": $4! = 24$.
- Gold, silver, bronze among 12 students: $(12)_3 = 1320$.
- 6 balls into 4 boxes: $4^6 = 4096$ if the balls are distinct, but $\left(\!\binom{4}{6}\!\right) = \binom{9}{3} = 84$ if they are identical. Distinguishability matters!
- 5 distinct colors on 5 chairs in a row: $5! = 120$.
- Passwords of length n: 26^n.
- 7 tasks, 3 employees, each gets at least one task: 1806. This needs the **Inclusion–Exclusion Principle** (next chapter).

Thank you!

Questions?

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